A
PROJECT REPORT
on

"IOT BASED SOLAR CROP DRYING"

Submitted to:
Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur for Partial Fulfillment of the Degree of

Bachelor of Technology

Submitted by

Mr. Vedant Nanoti
20241027220818

Mr. Sarang Kachare
20241027220809

Mr. Rohan Tonge
20241027220800

Under the Guidance of
Dr. Abhimanyu Dutonde



Department of CSE-Data Science

NAAC Accredited with A+ Grade

Vidarbha Bahu-uddeshiya Shikshan Sanstha's

Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur-441108

(An Autonomous Institute affiliated to RTMNU, Nagpur)

Session 2026-27

Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur-441108

(An Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

CERTIFICATE

This is to certify that project work described in this report entitled, "**IOT BASED SOLAR CROP DRYING**" was carried out by **Mr. Vedant Nanoti, Mr. Sarang Kachare, Mr. Rohan Tonge,** in Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur under my supervision and guidance in partial fulfillment of the requirement for the degree of **B.Tech CSE-Data Science** of Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur.

This work is the own work of the candidate, completed in all respects and is of sufficiently high standard to warrant its submission to the said degree. The assistance and resources used for this work are duly acknowledged.

Dr. Abhimanyu Dutonde
Guide

Prof. Sayara Bano Sheikh
HoD,(CSE-DS)

Dr.Dipali Bhoyar
Dean Academics, UG

Dr. Ganesh Wakte
Vice Principal

Dr. Pragati Patil
Director Administration

Dr. P.L. Naktode
Principal

Signature of External Examiner

Signature of Internal Examiner

Date: / / 2026





Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur-441108



(An Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

DECLARATION

I hereby declare that this Project titled **"IOT BASED SOLAR CROP DRYING"** is a bonafide and authentic record of the work done by me under supervision of **Dr. Abhimanyu Dutonde** during academic session 2026-27.

The work presented here is not duplicated from any other source and also not submitted earlier for any other degree/diploma of any university. I understand that any such duplication is liable to be punished in accordance with the university rules. The source material, data used in this research have been duly acknowledged.

Date:

Place:

Name and Signature of Students

Mr. Vedant Nanoti

Mr. Sarang Kachare

Mr. Rohan Tonge





Tulsiramji Gaikwad-Patil College of Engineering & Technology, Nagpur-441108



(An Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

ACKNOWLEDGEMENT

With profound feeling of immense gratitude and affection, I would like to thank my guide **Dr. Abhimanyu Dutonde**, **Assistant Professor, Department of CSE-Data Science**, for his continuous support, motivation, enthusiasm and guidance. His encouragement, supervision with constructive criticism and confidence enabled me to complete this project.

I, *also wish to extend my reverences to the* ***Dr. Abhimanyu Dutonde*** *for valuable support and technical insight to complete this project.*

I also wish to extend my reverences to **Prof. Sayara Bano Sheikh, Department of CSE-Data Science** for providing necessary facilities to complete my project.

I am also thankful to all the faculty members and all non-teaching staff of the department & college for their cooperation throughout the project work.

I also put forth my/our deepest sense of gratitude towards the **Dr. Pragati Patil,** Director, Administration and **Dr. P. L. Naktode,** Principal, TGPCET for constant motivation and providing necessary infrastructure.

Project Members

Date:

Place:

Mr. Vedant Nanoti
Mr. Sarang Kachare
Mr. Rohan Tonge

PLAGIARISM REPORT OF THIS WORK

Ms.Ashwini Sabal

IOT Based Solar Crop Dryer.docx

A105

Document Details

Submission ID
trn:oid:::18453:152685779

Submission Date
Sep 23, 2026, 3:20 PM GMT+5:30

Download Date
Sep 23, 2026, 3:22 PM GMT+5:30

File Name
IOT Based Solar Crop Dryer.docx

File Size
2.6 MB

92 Pages

19,932 Words

113,717 Characters

6% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

104 Not Cited or Quoted 6%
Matches with neither in-text citation nor quotation marks

0 Missing Quotations 0%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

3% Internet sources
1% Publications
5% Submitted works (Student Papers)

Certificate(s)

CONTENTS

Certificate i
Declaration ii
Acknowledgement iii
Plagiarism Report on the present work iv
Certificate(s) for Project Work v
Contents ..... vi-ix
Table Index x
Figure Index xi
Abstract xii

CHAPTER – I

INTRODUCTION 1-8
1.1 Summary 1
1.2 Background 2
1.3 Problem Statement 3
1.4 Aim and Objective 4
1.5 Scope 5
1.6 Design specification 6
1.7 Chapter Summary 8

CHAPTER – II

Literature Review and Technology Background 9-17
2.1 Solar dryer classification 9
2.2 Hybrid operation 10
2.3 Quality consideration 11
2.4 Importance for tray and airflow design 12
2.5 Literature comparison 13
2.6 Related IOT crop dryer 14
2.7 Research Gap 15
2.8 Contribution 16
2.9 Chapter Summary 17

CHAPTER – III

System Requirements and Design Methodology 19-25

3.1 Introduction……19

3.2 System Requirements 20

3.2.1 Function Requirement……...21

3.2.2 Non-functional Requirements……...22

3.3 Design Methodology……..23

3.4 Phase 1: Structure and core electronics……....23

3.5 Phase 2: Intelligence, power upgrade, and remote monitoring……..24

3.6 Phase 3: Finishing, safety and validation……..……25

3.7Transformation Analysis..... 25

CHAPTER – IV

CROP SELECTION AND DRYING REQUIREMENTS……..26-33

4.1 Introduction ……27

4.2 Selection of Target Crop ……28

4.3 Physical Characteristics of Selected Crops…….29

4.4 Initial Moisture Content ……30

4.5 Recommended Drying Temperature ……31

CHAPTER – V

MECHANICAL DESIGN OF THE SMART CROP DRYER……..34-40

5.1 Introduction ……34

5.2 Overall Mechanical Design ……36

5.3 Design of The Drying Chamber ……38

5.4 Material Selection ……..40

CHAPTER – VI

Electrical and Electronic System Design 41-53

6.1 Introduction 41

6 **2** Electrical System Architecture 42

6.3 Microcontroller Selection……..43

6.4 Power Supply Design..... 44
6.5 Heating Element……..……..45
6.6 Fan/Blower System……46
6.7 Temperature Sensor……...47
6.8 Humidity Sensor……48
6.9 Moisture Monitoring Approach……49
6.10 Weight Sensor/Load Cell……49
6.11 Relay and Switching Circuit……50
6.12 Motor Driver……51
6.13 Sensor Placement……51
6.14 Electrical Circuit Diagram……..52
6.15 Power Consumption……52
6.16 Electrical Safety……53
6.17 Hardware Integration……53

CHAPTER – VII

ELECTRICAL AND ELECTRONIC SYSTEM DESIGN……54-63

7.1 Introduction 54
7.2 Objectives of the IoT Architecture ……55
7.3 Overall System Architecture …… 56
7.4 Layered IoT Architecture ……..57
7.4.1 Physical / sensor layer ……57
7.4.2 Processing layer …… 58
7.4.3 Communication layer …… 58
7.4.4 Application layer …… 59
7.5 Sensor Layer……..59
7.6 Temperature Sensor……...59
7.6.1 Humidity Sensor……..60
7.7 Microcontroller Architecture……60
7.8 Actuator Layer……61
7.9 Heating System ……61
7.10 Ventilation Fan ……62
7.11 IoT Communication Architecture……63
7.11 Wi-Fi Communication……..63

CHAPTER – VIII

Datasets and Experimental Methodology 64-72

8.1 Introduction 64

8.2 Evaluation Datasets 65

8.3 Data Preparation and Preprocessing 66

8.4 Experimental Protocol 67

8.5 Evaluation Metrics 68

8.6 Experimental Configurations 69

8.7 Baselines and Comparative Evaluation 70

8.8 Chapter Summary 72

CHAPTER – IX

Results and Performance Analysis……72-79

9.1 Introduction 72

9.2 Overall Performance of EHSA 72

9.3 Dataset-wise Performance Analysis 72

9.3.1 EHSA-Synth-180 73

9.3.2 IBM CodeNet-100 73

9.3.3 TransBench-Lite-210 73

9.3.4 OJClone-32 74

9.4 Comparison of Individual Similarity Channels 74

9.5 Confidence-Interval Analysis 74

9.6 Confusion Matrix Analysis 75

9.7 Analysis of Hard Negatives 75

9.8 Ablation Study 75

9.9 Adaptive Fusion Results 76

9.10 Statistical Significance Analysis 76

9.11 Overall Discussion 77

9.12 Chapter Summary 79

REFERENCES 80

Research Paper / Publication .....

TABLE INDEX

Sr. No.	Table No.	Name of Table	Page No.
1	Table 1.1	Scope of the hardware design and development	3
2	Table 1.2	Design and performance parameters of solar dryer	6
3	Table 2.1	Classification matrix of the solar dryers	9
4	Table 2.2	Target drying parameters by product	10
5	Table 2.3	Literature review and design implications	11
6	Table 2.4	Adopted design parameters of the solar crop dryer	13
7	Table 3.1	List the complete Phase 1 bill of materials as procured	14
8	Table 6.1	Electrical and Electronic component specifications	16

FIGURE INDEX

Sr. No.	Figure No.	Name of Figure	Page No.
1	Fig. 5.1	General arrangement of the proposed solar batch crop dryer	01
2	Fig. 6.1	System block diagram of the IOT Solar Crop Dryer	14
3	Fig. 6.2	Power and Control wiring schematic	15

ABSTRACT

The integration of IoT, data analytics, and intelligent monitoring has become increasingly important in the development of modern agricultural systems. Software plays a crucial role in enabling real-time monitoring, data communication, automated control, and decision-making in smart crop-drying applications. This thesis focuses on the software development of a Smart Portable Multi-Crop Dryer, designed to monitor and manage the drying process using sensor-based data and an IoT-enabled architecture. The proposed software system collects temperature, humidity, and weight-related data from the drying chamber and processes this information to provide continuous monitoring of crop-drying conditions.

The proposed system integrates an ESP32-based sensing unit with a Raspberry Pi for data processing and communication. MQTT is used for reliable transmission of sensor data, while Node-RED provides data processing and control-flow management. Influx DB is used for storing time-series data, and Grafana provides an interactive dashboard for real-time visualization of temperature, humidity, weight, drying status, and drying progress. The software architecture is designed to support automated monitoring and control, reducing the need for continuous manual supervision and providing users with accessible information about the drying process.

The proposed software solution aims to improve the efficiency, reliability, and transparency of crop drying by combining real-time sensing, data visualization, and automated decision-making. Continuous monitoring can help maintain suitable drying conditions and reduce the possibility of over-drying or under-drying. The system also provides a foundation for future integration of drying-time prediction, moisture forecasting, data analytics, remote monitoring, and intelligent control techniques. The study demonstrates how IoT and data-driven software technologies can contribute to the development of a more efficient, sustainable, and user-oriented agricultural drying system.

CHAPTER – I

INTRODUCTION

CHAPTER - I

INTRODUCTION

1.1 Summary:

The proposed system integrates an ESP32-based sensing unit with a Raspberry Pi for data processing and communication. MQTT is used for reliable transmission of sensor data, while Node-RED provides data processing and control-flow management. Influx DB is used for storing time-series data, and Grafana provides an interactive dashboard for real-time visualization of temperature, humidity, weight, drying status, and drying progress. The proposed software solution aims to improve the efficiency, reliability, and transparency of crop drying by combining real-time sensing, data visualization, and automated decision-making. Continuous monitoring can help maintain suitable drying conditions and reduce the possibility of over-drying or under-drying. The system also provides a foundation for future integration of drying-time prediction, moisture forecasting, data analytics, remote monitoring, and intelligent control techniques

1.2 Background

Drying is the oldest and still the most widely practiced method of preserving agricultural and plantation produce. Removing water lowers the water activity of the product below the threshold at which moulds, yeasts, bacteria and enzymes remain active, so that a correctly dried crop can be stored for months without refrigeration. Safe storage moisture content is crop-specific: it is commonly quoted as 12-14% (wet basis) for cereals, 8-10% for chillies and other spices, 9-12% for pulses, and roughly 10-12% for cured crop.

Post-harvest losses of cereals, pulses and plantation crops in South Asia are estimated at 6-12% of production, a large share of which arises from inadequate or uncontrolled drying, storage-stage fungal growth and, in the worst case, aflatoxin formation. Reducing this loss is a stated national priority, and low-cost, farmer-operable drying technology is repeatedly identified in the literature as the most direct lever available to smallholders and primary processing co-operatives.

This traditional open-sun process has four well-documented disadvantages that are common

Open-sun drying of any crop: (i) it is entirely weather-dependent, and an unseasonal shower during the harvest window can spoil or discolour an entire batch; (ii) the product is exposed to dust, insects, rodents and bird droppings; (iii) the drying temperature and airflow cannot be controlled, so the drying

rate varies with time of day and with tray position, producing non-uniform colour and case-hardening; and (iv) the long duration ties up drying yards, labour and working capital for well over a month per batch, during which the grower or society is fully exposed to price risk.

Mechanical hot-air dryers exist commercially for crops and remove the weather dependency, but they typically draw 1.5-3 kw of grid or diesel-generator power continuously for many hours, which is not a viable proposition for a smallholder or a small cooperative operating on intermittent rural feeders. Solar thermal energy is attractive in this context precisely because the crop harvest seasons coincide with periods of comparatively high insolation, and because the sir temperature required for grain and crop drying – broadly 45-60 °C is comfortably within reach of a simple, low-cost flat-plate air collector fabricated from locally available sheet metal, glazing and insulation.

A purely solar dryer, however, is itself weather-dependent and gives no output overnight or on heavily overcast days, which reintroduces the very uncertainty that mechanization was meant to remove. The present project therefore adopts a hybrid architecture: a solar air collector supplies the shortfall on low-insolation days so that a batch started in the morning can still be completed within the working day, and a WiFi-capable microcontroller with calibrated temperature, humidity and electrical sensors gives the operator (and in the complete project, a cloud dashboard) continuous visibility of the drying process – an IOT-ready instrumentation layer that a purely solar or purely mechanical dryer does not offer.

The IOT dimension of the project is deliberately treated as an instrumentation and monitoring layer rather than as an autonomous control layer: sensors record the drying-air temperature, relative humidity, grain/crops bed temperature and photovoltaic/battery electrical quantities at fixed location and a fixed interval, and this record is what allows the collector efficiency, system efficiency, drying rate and specific enery consumption claimed in the abstract to be computed after the fact from real data rather than assumed. This mirrors the emphasis of the pre-submission hardware audit that a project claiming “IOT” operation must be able to show a defined sensor set, a defined interface, and a resulting dataset- not merely a WiFi-connected microcontroller.

1.3 Problem Statement

A dryer intended for smallholder or co-operative-society use must simultaneously satisfy four requirements that are usually in tension with one another: it must be affordable to fabricate from locally available material; it must not depend on a continuous, reliable grid supply; it must hold the crop within a temperature band that does not damage colour, germination or milling/market quality; and it must finish a batch fast enough that the operator is not tempted to revert to open-sun drying to

free up tray space. Purely passive solar cabinet dryers satisfy the first two requirements but fail the last two, because natural convection gives an uncontrolled, low airflow and the cabinet air temperature can exceed 70°C at solar noon while collapsing back to near-afternoon or under cloud.

For crop specifically, the problem is sharpened by the sheer duration of the conventional process. An open-sun cycle of twenty to forty-five days represents weeks of tray-yard occupancy, labour for daily turning in drying time, combined with protection from rain and contamination, therefore has a disproportionate benefit relative to its cost.

A separate, documentation-level problem motivates the present report. A pre-submission hardware audit of an earlier draft of this project found that the original document asserted "solar" and "IOT" operation without stating hardware/software boundary, without any dimensioned mechanical drawing, without stating a heater or collector sizing calculation, without any airflow or pressure-drop figures, without electrical protection or earthing detail, without a costed bill of materials, and without any measured result – the report was written entirely in an instruction verification matrix of are structured specifically to close these documentation gaps by fixing an explicit scope, a consolidated design specification, and a traceable path from every requirement to a named verification method to be exercised in the engineering-design, fabrication and experimental chapters that follow this report.

A further, crop-specific dimension of the problem is that crop processing is almost always undertaken by primary producers and small co-operative societies operating on thin margins, so that any proposed mechanized alternative must be justified on a simple, transparent cost basis-a stated material-cost target of table in the costed bill of materials developed in Chapter 5 of the complete thesis.

The problem addressed by this thesis can therefore be stated concisely as: to design a protected, forced-convection, hybrid solar dryer for a 20kg-class wet batch that removes the batch to a safe storage moisture content within a single working day, holds the crop below its damage temperature at all times, draws its control and sensing electronics entries from a stand-alone photovoltaic supply, and is documented to an auditable engineering standard-dimensional, calculated, costed and later experimentally verified-rather than described only in qualitative term.

1.4 Aim and Objectives

Aim: to design, and to document to an auditable engineering standard, a solar-powered, IOT-instrumented, forced-convection hybrid batch dryer that is applicable to crop and to reference grain crop, that can reduce a 20 kg-class wet batch to a safe storage moisture content within a single working

day, and that holds the crop below its safe drying temperature at all times through an independent, hardware-backed protection scheme.

The specific objectives of the work reported in carried forward into the engineering-desgin chapters of the complete thesis, are:

1. To review the classification of solar dryers and the published literature on hybrid, indirect, forced-convection dryers, and to fix the design parameters (drying temperature band, target collector and system efficiencies, superficial air velocity, maximum bed depth) adopted for this project.

2. To review at least one published IOT-based crop dryer in detail, and to identify from that review and from the wider literature, the specific research gap that this project addresses.

3. To establish the drying load-the mass of water to be removed, the useful energy and the input energy- from first principles for the design crop and batch size, as the basis for the sizing calculations .

4. To size the flat-plate solar air collector. The drying chamber, the tray system and the air-circulation path from the calculated thermal and airflow requirements.

5. To design a stand-alone photovoltaic supply, with battery back-up and correctly rated electrical protection, for the fans, controller, sensors and relays, independent of the auxiliary heater load.

6. To specify a two-level, hardware-backed over-temperature protection scheme and to define how it will be verified experimentally.

7. To define the instrumentation calibrated temperature, humidity, bed-temperature and electrical sensors at defined, justified locations, and to specify the IoT-ready interface through which they will be read.

8. To fabricate the complete unit from locally available materials within a stated target material cost, and to document the fabrication sequence, jigs, joints and finishes in the later chapters of the complete thesis.

9. To define a functional and non-functional requirement set and a requirement verification matrix that traces every design target to a named verification method, so that every claim made in this report can later be checked against a physical measurement.

10. To define an experimental protocol covering at least three full batches compared against an open-sun control for reporting collector efficiency, system efficiency, drying rate, specific energy consumption and product-quality outcome in the results chapters of the complete thesis.

1.5 Scope

This report is a hardware engineering report. Its scope, and the corresponding exclusions, are fixed explicitly the hardware/software boundary a point specifically raised by the pre-submission audit is unambiguous from the first chapter onward.

table 1.1 scope of the hardware design and development

Included in scope (hardware)	**Explicitly excluded**
Thermal, airflow, structural, solar and electrical design calculations	Firmware source code and its software architecture
Mechanical fabrication, joining, insulation and finishing	Mobile / web dashboard user-interface design
Component selection, ratings, bill of materials and costing	Cloud database schema and API design
Sensor selection, placement, wiring and calibration procedure	Machine-learning or predictive-analytics claims
Electrical protection, earthing and safety verification	Economic modelling beyond a simple payback estimate
Experimental testing and performance characterisation	Crop biochemistry and nutrient-retention assays
Areca-nut and reference-grain (paddy) drying trials	Large-scale continuous or industrial dryer configurations

Delimitations: the design is developed for a 20 kg-class wet batch and is intended to be prototyped and tested as a reference/control crop, at a single site during a single drying season. Results obtained from the prototype should not be extrapolated to oil-bearing seeds, to batch sizes materially larger than 20kg, or to ambient and insolation conditions markedly different from those recorded during the trails described in the experimental chapters of the complete thesis.

1.6 Design Specification

The quantitative design values used throughout this report are adopted from a validated model/reference thesis for a 5 kg-capacity, indirect, forced-convection, hybrid solar batch dryer, and are combined with the corresponding crop specific parameters drawn from the literature reviewed in every value in marked with a status flag DESGIN values are calculated design targets carried forward unchanged from the reference thesis VERIFY values are component ratings that must be confirmed against the manufacturer data sheet of the actual part procured MEASURE values must be replaced by a physical measurement taken on the fabricated prototype and test values must be replaced by the outcomes of a defined commissioning or safety test.

Ambient design conditions representative of a midmorning to midafternoon drying window at the project site (design insolation of the order of 700 W/m² on the collector plane, and an ambient temperature of the order of 30 °C), and assumes that the auxiliary heater is sized to make good the thermal shortfall only — it is not sized to dry the batch unassisted. These assumptions are carried forward into the load and collector-sizing calculations of Chapter 4 of the complete thesis and must be re-checked against the recorded weather data of the actual trial period before the design is finalized.

table 1.2 Design and performance parameters of solar dryer

Parameter	Reference / design value	Status
Batch capacity	5 kg wet crop (paddy reference; areca nut target)	MEASURE
Initial moisture content (paddy reference)	25% w.b.	MEASURE
Final moisture content (paddy reference)	13% w.b.	MEASURE
Initial moisture content (crops boiled)	≈ 45–55% w.b. (crop- and variety-dependent)	MEASURE
Final moisture content (crops)	≈ 10–12% w.b.	MEASURE
Drying air temperature at tray inlet	50–55 °C; design maximum 60 °C	MEASURE

Parameter	Reference / design value	Status
Firmware over-temperature set-point	55 °C	TEST
Independent hardware over-temperature cut-out	65 °C nominal, wired in series with heater contactor	TEST
Auxiliary PTC heater rating	500 W	VERIFY
Solar air collector aperture area	2.0 m², flat-plate, matt-black absorber	MEASURE
Collector thermal output at design insolation	≈ 630 W (design)	DESIGN
Design airflow rate	≈ 138 m³/h	MEASURE
Fans	2 × 120 mm axial, 12 V DC	VERIFY
Superficial air velocity through the tray bed	0.10–0.30 m/s	MEASURE
Maximum grain	≤ 80 mm	MEASURE
Number of trays	3, perforated, removable	MEASURE
Photovoltaic supply	50 Wp	VERIFY
Battery	12 V, 26 Ah, sealed lead-acid, 1-night autonomy	VERIFY
Low-voltage electronics load (design)	≈ 14.5 W (controller + sensors + relays)	MEASURE
Drying chamber internal envelope	0.50 m × 0.50 m × 0.60 m (design)	MEASURE
Insulation	25 mm rigid PU foam or 50 mm glass wool	VERIFY
Controller	ESP32 (Wi-Fi / Bluetooth capable microcontroller)	VERIFY

Parameter	Reference / design value	Status
Sensing	Temperature/RH at collector outlet, plenum, mid-tray and exhaust; bed-temperature probes; PV/battery voltage and current	VERIFY
Target batch duration	≤ 8 h per batch (reference grain); multi-day for crops, to be reduced relative to open-sun control	TEST
Material cost target	≤ INR 50,000 (reference)	MEASURE

1.7 Chapter Summary

The post-harvest drying problem in general and for crop in particular stated the aim and ten supporting objectives, fixed the hardware/software boundary of the work, and consolidated the reference design specification with an status flag against every parameter Chapter 2 now reviews the literature that justifies these design choices in detail.

CHAPTER – II

LITERATURE REVIEW AND TECHNOLOGY BACKGROUND

CHAPTER - II

LITERATURE REVIEW AND TECHNOLOGY BACKGROUND

2.1 Solar Dryer Classification

Solar dryers are conventionally classified along three independent axes. The first is the mode of air movement natural convection dryers rely on buoyancy alone and therefore give an airflow that with the instantaneous temperature difference between the dryers and the ambient air, while forced convection dryers use a fan or blower to impose a controllable, repeatable airflow independent of that temperature difference. The second axis is the path taken by solar energy to reach the product in a direct dryer the crop itself is exposed to solar radiation inside a transparent enclosure, which is inexpensive but causes surface bleaching, vitamin loss and for indirect dryer air is heated in a separate opaque or glazed collector and then passed over the crop inside a shaded chamber, giving better colour uniformity and temperature control at the cost of a large, more complex structure a mixed-mode dryer combines both paths. The third axis is the presence or absence of auxiliary heating a solar only dryer has zero running cost but zero output after dark or under heavy cloud, while a hybrid dryer add a small auxiliary heat source electrical biomass or otherwise to make good the shortfall.

table 2.1 Classification matrix of the solar dryers

Classification axis	Categories	Relevance to this project
Air movement	Natural convection; Forced convection	Forced convection selected gives a repeatable, controllable airflow independent of ambient conditions
Solar-energy path	Direct; Indirect; Mixed-mode	Indirect selected better colour uniformity for areca nut and grain, and permits an enclosed, protected chamber
Auxiliary heating	Solar-only; Hybrid	Hybrid (limited electric) selected removes the risk of a batch spanning more than one working day

Classification axis	Categories	Relevance to this project
Operating mode	Batch Continuous/tunnel	Batch selected matches smallholder/co-operative-society scale and tray-based loading practice

The unit designed in this project is therefore an indirect, forced-convection, hybrid, batch dryer. This configuration was selected because grain and nut quality, temperature control and same-day (or substantially shortened) batch completion were all ranked above minimum first cost in the objective list to fail one or more of these requirements under realistic weather conditions.

2.2 Hybrid Operation

Solar-only drying is strongly, and unavoidably dependent on instantaneous irradiance output falls to zero at night and drops sharply under cloud so that a solar-only dryer sized for the design insolation will under-perform on a below average day and may leave a batch partially dried overnight risking mould growth and reversing the very quality benefit the dryer was built to provide. A limited auxiliary heater restores continuity during cloudy or late-day conditions without requiring the collector fan and duct system to be oversized for the worst-case day which would otherwise inflate the first cost of the unit.

A key design decision, adopted directly from the reference model thesis, is to keep the auxiliary heater outside the photovoltaic sizing calculation a 500 W resistive heating element run from a solar-electric supply would require a photovoltaic array and battery bank an order of magnitude larger and correspondingly more expensive than one sized only for the low-power fans, controller, sensors and relays. The auxiliary heater is therefore supplied from a conventional single-phase electrical connection (or, where unavailable, an equivalently rated small generator), protected by its own miniature circuit breaker and residual current device while the photovoltaic subsystem is reserved exclusively for the low-voltage instrumentation and control load. This split is what allows the unit to be described, without overstatement, as "solar" for its thermal energy and its control electronics, while remaining transparent about the origin of the auxiliary heat a distinction the pre-submission audit specifically required to be made explicit.

The fraction of total drying energy supplied by the sun (the "solar fraction") is quantified experimentally in the results chapters of the complete thesis by metering the heater's on-time against the total batch duration; a solar fraction target of at least 70% of total thermal energy (equivalently, an

auxiliary-heater duty of no more than 30% of total thermal energy) is carried into the requirement verification matrix as requirement R6. Thermal-storage alternatives to an electric auxiliary heater a rock-bed store or a phase-change-material store charged during peak insolation and discharged in the late afternoon were considered and are noted as a direction for future work rather than as part of the present scope, because they add significant volume and cost for a batch-duration benefit that a small electric heater achieves more simply.

2.3 Quality Considerations

Controlled temperature and airflow reduce the risk of three quality defects that are well documented for both grain and plantation crop drying over under drying (residual moisture above the safe storage threshold, leading to mould and aflatoxin risk during storage), and case hardening. The protected, opaque chamber of an indirect dryer additionally removes the ultraviolet bleaching and dust/insect contamination risk that a direct or open sun process carries.

For crop is the dominant commercial quality criterion and colour development is governed largely by the rate of moisture loss during the first days of drying immediately after boiling too rapid a drying rate produces a dark unevenly coloured crop that is downgraded at the point of sale while too slow a rate extends the exposure window during which fungal spoilage can begin. This is the technical justification additional to the general energy cost argument of for holding the drying air temperature within a controlled band (50–55 °C, absolute maximum 60 °C for the reference grain; the corresponding crop specific band is to be confirmed by trial and cross-checked against the published range) rather than allowing the uncontrolled swings of a passive solar cabinet.

table 2.2 Target drying parameters by product

Product	**Typical initial moisture**	**Target final moisture**	**Drying-air temperature limit**	**Dominant quality risk**
Paddy (reference/control crop)	22–28% w.b.	12–14% w.b.	≤ 60 °C (milling paddy); ≤ 43 °C (seed paddy)	Case-hardening; internal cracking; germination loss (seed use)
Crops, boiled (target crop)	≈ 45–55% w.b.	≈ 10–12% w.b.	To be fixed by trial; literature and related-work range ≈ 45–60 °C	Surface cracking; mould if under-dried

Product	Typical initial moisture	Target final moisture	Drying-air temperature limit	Dominant quality risk
Chilli (literature comparator)	≈ 70–75% w.b.	≈ 8–10% w.b.	≤ 60 °C	Colour/pungency loss at excess temperature

2.3 Implications for tray and airflow design.

The quality considerations above translate directly into two design constraints carried into Chapter 3 and into the engineering design chapter of the complete of the complete thesis. First because case hardening and colour non-uniformity are both driven by an uneven temperature and humidity filed across the stack the bed depth is capped at 80 mm and the superficial air velocity is held within a 0.10-0.30 m/s band so that every layer of the bed receives comparable drying air rather than the top layer drying quickly while the bottom layer remains wet. Second because crop is processed as a batch that is turned periodically by hand in the traditional process the tray system is required to be tool-free and removable so that the same manual turning mechanism that would add cost and complexity outside the present scope. A secondary, but commercially important quality consideration for crop is weight loss during storage after drying crop dried to a lower than moisture content loses saleable weight without any corresponding quality benefit while crop left above the safe threshold risks mould driven weight and grade loss during storage. This is the underlying reason the final moisture target in expressed as a band (10-12% w.b) rather than a single value and why the moisture reduction requirement is verified by an oven method determination rather than by a single hand moisture meter reading which is known to be less reliable for high produce such as crop.

2.4 Literature Comparison

The principal sources consulted the key quantitative finding of each and the specific design implication adopted in this project. The table follows the same reference numbering resolves to a single listed source corresponding the pre-submission audit specifically required.

table 2.3 Literature review and design implications

Ref.	Author/year	System/topic studied	Key finding	Design implication adopted
1.	ASHRAE Handbook Fundamentals (2021)	Psychrometrics air properties	c_pa=1.0005kj/kg-k;h_fg=2382kj/kg at 50°C	Property values used in ch.4
2.	Elwakeel & Tantawy(2023)	PV-integrated solar dryer with iot based mointoring	Shows the automatic sensor controlled moisture content faster	IOT control loop moisture
3.	Review of technological development in solar dryers (2016-2023)	Multi-year review of solar dryer advances	Covers research contributions to solar dryers globallly	2022-2023 development and a specific future
4.	Review of smart solar dryer 2022 (IJRASET)	Status review of smart solar dryer (SSD)	Highlights that PID control system in smart solar	Your own control system choice
5.	ELLIS Opoku, Sekyere & arkoh (2023)	Solar crop dryer with thermal energy storage as backup heater	Relevant if you want to discuss thermal storage as an alternative to electric auxiliary heater	The phase change material (PCM) for specification

The hardware audit additionally recommended broader coverage pf PTC element behaviour, insulation selection tray arrangement photovoltaic sizing method and drying rate models than is available in a single review paper.

Table 2.4 Adopted design parameters of the solar crop dryer

Parameter	Adopted value
Drying air temperature at tray inlet	50-60 °C
Absolute over-temperature cut out	65 °C
Collector thermal efficiency	45%
Overall system drying efficiency	35%
Latent heat of vaporization at 50 °C	2382 KJ/kg
Specific heat of paddy	3.0kj/kg
Superficial air velocity through bed	0.10-0.30m/s
Maximum chillis/crop	<80 mm

2.5 Related IOT crop dryer

A published IJERT paper on an IOT based crop dryer describes a steel drying chamber fitted with electric heating elements, temperature sensing and Arduino based thermostatic control loop and additionally proposes a motorized scissor lift mechanism that raises and lowers the chamber to expose it to direct sunlight when insolation is favourable, combining solar exposure mode with an electrically heated mode within the same physical enclosure. The same paper outlines an esp32 based control architecture with temperature/humidity sensing-relay controlled heating and fan hardware wireless(Wi-Fi) communication and MQTT based publication of sensor data to a cloud broker for remote monitoring.

2.5.1 Architecture summary of the reviewed work

- A welded steel chamber housing the crop and more resistive heating elements with a mechanically actuated lifting frame for solar exposure.
- Point temperature sensing feeding a simple on/off Arduino thermostat loop.
- A separate ESP32 based variant with Wi-Fi connectivity relay driven heater and fan control and MQTT publication of reading to a cloud broker.

- No published mechanical drawing set, airflow figure, collector sizing calculation, photovoltaic sizing, bill of materials or independent hardware level over temperature protection distinct from the firmware thermostat.

2.5.2 Limitations identified relative to the present project

- The heating source in the reviewed work is presented as either gird-electric or direct solar exposure thermal output no collector aperture area absorber specification or design efficiency is reported.

- Temperature control relies on a single firmware thermostat loop with no stated independent hardware level cut out a firmware fault or sensor failure has no second line of defence against overheating.

- No airflow rate fan static pressure curve or pressure drop through bed figure is reported so the forced convection performance of the design cannot be independently checked.

- No photovoltaic or battery sizing is presented for the sensing/control electronics so the extent to which the system is genuinely solar powered for its electronics is not established.

- No costed bill of materials or measured drying performance data is reported against which a farmer or co-operative society could evalute the economics of adoption

These concepts the ESP32/Wi-Fi/MQTT control architecture in particular are adopted in the present project only as a related architectural reference for the IOT hardware chapter of the complete thesis and are not treated as performance results for this project every quantitative claim made about the present dryers is separately calculated or measrured.

2.6 Research Gap

Synthesising Sections 2.1 through 2.5, and the specific findings of the pre-submission hardware audit, the following gaps are identified and are addressed by the contribution set out in Section

- Instrumentation granularity. Most reported low-cost dryers, including the areca-nut work reviewed in instrumented with only a handful of point sensors, so the internal temperature and humidity field and hence the true tray-to-tray non- uniformity that drives the colour-grade problem of never quantified.

- Unverifiable solar claim. Hybrid dryers commonly draw the auxiliary heater from the grid without presenting an energy balance that shows what fraction of total drying energy was actually

solar; the "solar" claim is therefore unverifiable, a gap this project closes through the metered solar-fraction requirement .

- No areca-nut-specific sizing model. Collector, airflow and thermal-load calculations in the literature are almost exclusively developed for cereals and a small number of spices; no publicly available worked calculation adapts these methods to the very different initial moisture content, bulk density and colour-sensitivity of boiled.

- Absent or single-layer safety architecture. The reviewed areca-nut work, and several of the grain-drying papers surveyed in describe only a firmware thermostat with no independent hardware-level trip, leaving no second line of defence against a software or sensor fault.

- No costed bill of materials with measured performance. Very few student-built or small-scale prototypes report a full, costed bill of materials together with measured specific energy consumption precisely the information a farmer or a co-operative extension agency needs to evaluate adoption.

- Weak requirement traceability. The pre-submission audit found no traceable path from a stated requirement to a physical dimension, a component rating or a verification test in the original document; this gap is closed directly by the requirement verification matrix.

- No open-sun comparative benchmark. Several reviewed papers report an absolute drying time or efficiency without a simultaneously run open-sun control batch. making the claimed time or quality benefit difficult to attribute to the technology rather than to that day's weather.

2.7 Contribution

Against the gaps identified in Section 2.6, this project makes the following specific contributions:

- An engineering-report structure in which every quantitative claim is either a cited literature value, a worked design calculation, or a measured result explicitly labelled as such never an unlabelled assertion.

- A worked, first-principles design-calculation chain (moisture load → thermal load → collector sizing → airflow → tray/chamber sizing → electrical load → PV/battery sizing → structural check) adapted specifically for a 20 kg-class batch applicable to both paddy and areca nut.

- A dedicated, item review of a published IoT-based areca-nut dryer (Section 2.5), used to define the specific gap this project fills rather than merely cited as background.

- An explicit two-level, hardware-backed over-temperature protection scheme (firmware set-point plus an independent electromechanical cut-out) specified and scheduled for experimental verification, rather than a firmware-only thermostat.

- A photovoltaic supply sized independently of the auxiliary-heater load, with a metered solar-fraction requirement (R6) that makes the "solar-powered electronics" and "hybrid thermal" claims independently checkable.

- A component-level, costed bill of materials presented against a stated target material cost, so that the economics of the design can be evaluated by a prospective adopter.

- A requirement verification matrix that traces every design target listed in named verification method, closing the traceability gap identified by the pre-submission audit.

- A defined experimental protocol, including a simultaneously run open-sun control batch, so that any measured improvement in drying time or product quality can be attributed to the dryer rather than to that day's weather.

The underlying technologies solar air heating, forced-convection batch drying and ESP32-based IoT monitoring are individually well established in the literature reviewed above. The specific contribution of this project is the integrated, areca-nut-applicable hardware prototype, its auditable design documentation, and its experimentally verified performance dataset, rather than the invention of a new drying principle.

2.8 Chapter summary

Chapter 2 has classified solar dryers along three independent axes, explained the rationale for hybrid solar/electric operation, reviewed product-quality considerations for grain and areca nut, compared twelve literature sources against the design parameters adopted in this project, reviewed a published IoT-based areca-nut dryer in detail, and set out the seven-point research gap that the eight-point contribution is designed to close. Chapter 3 now turns these findings into an explicit, verifiable requirement set and design methodology

CHAPTER – III

SYSTEM REQUIREMENTS AND DESIGN METHODOLOGY

CHAPTER - III

SYSTEM REQUIREMENTS AND DESIGN METHODOLOGY

3.1 Introduction

This chapter present the system requirements and the design methodology adopted for the development of the solar batch crop dryer a 5kg capacity prototype intended for small-scale post-harvest drying applications. The design process was carried out in three distinct, sequentially funded phase each addressing a specific layer of the system the structural and core electronic layer (phase1) the intelligence and automation layer (phase2) and the finishing safety and validation layer (phase3). The phased methodology and safety compliant dryer within a constrained budget. The sections that follow describe the functional and non-functional requirements that shaped the design the rationale for the phased approach and a detailed enumeration of the components procured in each phase, along with their role in the overall system.

3.2 System Requirements

The requirements for the dryer were derived from the intended use case drying of approximately 5 kg of agricultural produce per batch using solar and electrically assisted heating and were classified into functional and non-functional.egukate

3.2.1 Functional Requirements

- The system hold and dry a batch of up to 5kg of crop material distributed across multiple perforated trays.
- The system shall regulate chamber temperature using a controlled heating element and forced air circulation.
- The system shall continuously measure temperature, humidity and sample weight to allow the drying end point to be determined.
- The system shall be capable of operating on solar power with battery backup, reducing dependence on grid electricity.
- The system provide local display and remote indication of drying status.

- The system shall be mobile and easy to reposition between the drying site and storage.

3.2.2 non-functional requirements

- Reliability the control and sensing electronics must operate continuously through multi-hour drying cycles without failure.

- Safety the system must include cost was constrained to remain within the sanctioned project and a manual emergency stop.

- Affordability the total system cost was constrained to remain within the sanctioned project funding necessitating a phased and prioritised procurement plan.

- Durability exposed structural and electrical components must be weatherproofed and corrosion resistant for outdoor use.

- Usability operation and calibration should be achievable by using commonly available spares.

- Maintainability components should be modular and replaceable using commonly available spares.

3.3 Design Methodology

Given the fixed initial funding oh Rs. 20,000 and the broader goal of producing a fully instrumented solar powered field-reply unit a phased design build test methodology was adopted rather than attempting to specify and procure the complete system at once. This approach reduced financial risk allowed each subsystem to be validated before the next layer was added, and ensured that the most basic functional requirement the ability to physically hold and dry a 5 kg batch was satisfied first with intelligence connectivity and safety/finishing features layered on in subsequent phase as subsequent phases as further funding and design insight became available.

The methodology can be summarized as a three stage build sequence:

- Phase 1-Structural fabrication and core sensing/heating electronics.

- Phase 2- Upgrade to embedded computing solar-battery power management and remote monitoring.

- Phase 3- Finishing mechanical safety calibration and validation (field-readiness layer.)

Each phase concluded with a component level cost reconciliation against its allocated budget, which is discussed alongside the component list in the section below. This traceability between requirement,

component, and cost is a central feature of the design methodology, since it demonstrates that every rupee of the sanctioned funding was mapped to a specific functional need of the dryer.

3.4 Phase1: Structural and core Electronics

Phase 1 corresponding to the foundational build of the dryer and was funded from the initial project sanction of Rs. 20,000. The objective of this phase was to fabricate the physical drying chamber and equip it with the minimum set of sensing heating and control electronics needed to run a basic bench supervised drying cycle. The component list is dominated by structural fabrication items the MS/aluminium frame sheet metal chamber body perforated stainless steel trays glazing and insulation which together from the insulated solar collection and drying chamber. A large number of small fasteners and sealing items were required to assemble the chamber and ensure it remined weather.

Table 3.1 list the complete phase1 bill of materials as procured.

No.	Component name	Qty	unit	Amount(Rs.)
1.	MS/aluminium frame	1	Set	3,500
2.	Chamber body sheet metal	1	Sheet	1,500
3.	Perforated ss trays	3	Pc	2,000
4.	Polycarbonate glazing sheet	1	Sheet	1,200
5.	Insulation	1	Set	1,000
6.	Door hinges	2	Pc	150
7.	Door lock	1	Pc	100
8.	Nuts M4/M6	40	Pc	100
9.	Bolts	40	Pc	150
10.	Washers	40	Pc	60
11.	Rubber washers	15	Pc	100
12.	Spring washers	20	Pc	50
13.	Self-tapping	20	Pc	80
14.	Silicone sealant	1	Pc	200
15.	Rubber gasket strip	2	m	200
16.	L-clamps/brackets	6	Pc	180
17.	ESP32 dev board	1	Pc	400

18.	DHT22 sensor	1	Pc	200
19.	Load cell(5-10kg)	1	Pc	250
20.	HX711 amplifier module	1	Pc	100
21.	k-type thermocouple	1	Pc	200
22.	MAX6675 module	1	Pc	150
23.	PTC heater element	1	Pc	800
24.	Relay module	1	Pc	200
25.	Buck converter	1	Pc	100
26.	SMPS/adapter(12V)	1	Pc	500
27.	Jumper wires	3	Sets	150
28.	Breadboard	1	Pc	150
29.	Terminal blocks	5	Pc	150
30.	Heat shrink tubing	1	Pack	100
31.	Dc exhaust fan	2	pc	300
32.	Cable ties	1	Pack	60
33.	Insulated wire	5	M	150
34.	Fuse holder	1	Pc	80
35.	On/off switch	1	Pc	60
36.	Enclosure	1	Pc	250
37.	Double sided tape	1	Pc	80
38.	Solar panel (25w)	1	Pc	1,100
39.	MPPT charger controller	1	Pc	700
40.	SHT31 sensor	1	Pc	500
			Total	17,300

3.5 phase 2: Intelligence, power upgrade, and remote monitoring

With the structural prototype and basic control loop validated in Phase 1, Phase 2 focused on transforming the dryer from a bench-supervised device into a semi-autonomous, remotely monitorable unit capable of sustained solar-only operation. The centrepiece of this phase is the Raspberry Pi 4 (4GB), which was introduced as a higher-level supervisory computer to run data-logging, decision logic, and a monitoring interface alongside the existing ESP32 control loop. To support the additional computing load and enable meaningful off-grid runtime, the solar generation capacity was upgraded to a 40-60W panel paired with a full-specification MPPT controller, a 12V LiFePO4 battery pack, and a dedicated Battery Management System (BMS) to protect the battery from over-charge, over-discharge, and thermal abuse. A DC-DC converter was added specifically to step the 12V system rail down to the 5V required by the Raspberry Pi.

Remote monitoring capability was implemented using a GSM/SIM800L module, allowing status alerts and readings to be transmitted without dependence on local Wi-Fi infrastructure — an important consideration given the outdoor, off-grid deployment context. A local OLED/LCD (I2C) display was added for on-site status readout, while an RPi camera module was included to allow visual monitoring of the crop and chamber condition, with a 32GB microSD card provisioned for onboard data and image logging. Mechanically, a servo motor was introduced to allow automated vent or tray control, and a blower fan of improved specification was added to upgrade air circulation beyond the Phase 1 baseline. A waterproof enclosure was procured specifically to house the Raspberry Pi and battery pack, and cable glands, a junction box, and additional wiring/connectors were added to weatherproof the expanded electrical system.

3.6 phase 3: Finishing, safety and validation

Phase 3 addressed the requirements that make the dryer safe, durable, and ready for field deployment and demonstration, rather than adding new sensing or computing capability. Surface protection was achieved through powder coating/painting materials and an anti-corrosion coating applied to exposed metal surfaces, extending the service life of the chamber under outdoor conditions. Mobility and handling were improved with four caster wheels and two carry handles/grips, directly satisfying the usability requirement that the unit be easy to reposition.

Electrical and thermal safety were reinforced through a higher-rated circuit breaker/fuse, an emergency stop switch, and a thermostat providing an independent temperature safety cutoff in addition to the software-based control implemented on the ESP32 in Phase 1. A calibration weight set was procured to allow the load cell to be periodically verified against known reference masses,

ensuring the accuracy of the weight-loss-based dryness determination. A dedicated microSD module was added for ESP32-side data logging, complementing the Raspberry Pi-based logging introduced in Phase 2. The remaining items in this phase a final wiring harness/conduit, nameplate/labels/stickers, printed and bound user manual, a test sample crop for trial runs, and a contingency allocation for spares reflect the transition from a working prototype to a documented, testable, and presentable unit.

CHAPTER – IV

CROP SELECTION AND DRYING REQUIREMENTS

CHAPTER - IV

CROP SELECTION AND DRYING REQUIREMENTS

4.1 Introduction:

Crop drying is one of the most important post-harvest operations because freshly harvested agricultural crops generally contain moisture at the level that is unsuitable for prolonged storage. Excess moisture can lead to fungal growth, insect activity, unwanted biochemical reactions, loss of nutritional quality and reduction in market value. Therefore, the reduction of crop moisture to an appropriate safe-storage level is essential for maintaining crop quality and reducing post-harvest losses. Traditional sun drying is widely used by farmers because it requires relatively simple equipment and little or no direct energy input. However, its effectiveness depends strongly on environmental condition like sun temperature presence. Solar radiation, relative humidity, wind velocity, clouds cover and rainfall all lead to influence the drying process. During cloudy or humid weather, the available solar energy is reduced while the surrounding air may contain high moisture. That's why, drying becomes slow and may remain incomplete. Extended exposure of partially dried agricultural products to humid environmental conditions can increase the possibility of deterioration. Temperature above 40 °C could affect the quality as well as germination capacity of crop.

The proposed research is used to address this problem through the development of a smart solar-electric crop dryer. The system is designed for a capacity of approximately 5 kg and is intended to dry soybean and tur separately. The dryer combines solar energy with battery-supported electrical energy so that the drying process does not depend entirely on direct sunlight. A 24 V PTC heater provides controlled thermal energy, while a 24 V DC blower provides forced air circulation through the drying chamber. The main objective of the crop-selection and drying-requirement stage is to establish the technical requirements necessary for drying soybean and tur properly. The important parameters considered here include crop characteristics, initial moisture content, final moisture content, drying temperature, drying duration, relative humidity, airflow, crop loading, drying chamber requirements, energy requirements and overall drying performance. An important consideration in this research is that soybean and tur will not be treated as same drying materials. Each crop will be dry separately because the crops differ in their physical characteristics, moisture condition and response to temperature. The same dryer can therefore be used for both crops, but it will process them separately.

4.2 Selection of Target Crop:

The target crops selected for the proposed smart drying system are:

1)Soybean

2)Tur (pigeon pea)

These crops are selected because they are important oilseeds and require appropriate post-harvest moisture management for safe storage. The research is particularly deals with preventing avoidable post-harvesting by providing farmers with an alternative to traditional outdoor drying.

The selection of these two crops also allows the dryer to be evaluated with more than one type of agricultural product. This is useful for demonstrating that the system is not restricted to a single crop. At the same time, the crops will be tested independently so that the drying behavior of each crop can be studied separately with software integration later. The two crops will not be mixed during the drying process at all.

Soybean (*Glycine max*) , is an important agricultural crop containing substantial amounts of protein and oil. Because of its composition, soybean quality is sensitive to inappropriate post-harvest handling and storage conditions. High grain moisture can increase the risk of deterioration, while excessive drying temperature can affect physical and chemical quality. While studying we observed that temperature above 40°C can affect the quality and germination capacity of crop, drying at 40°C to approximately 12% moisture provided better preservation than combinations involving higher temperature and higher moisture. Soybean is also particularly relevant to the research objective of safe storage. Extension guidance commonly recommends drying soybeans to approximately 12–13% moisture for storage, with lower moisture requirements potentially applicable when longer storage or seed preservation is intended.

Tur, commonly known as pigeon pea or red gram, is another important pulse crop. Like soybean, it must be dried before storage to minimize moisture from it. Tur will be evaluated independently from soybean because the physical structure and composition of the grain are completely different. Differences in grain size, surface characteristics, moisture distribution, and thermal behaviour can influence the drying process. The use of tur as a second target crop also provides an opportunity to investigate whether the proposed dryer can maintain satisfactory drying conditions for different agricultural commodities without requiring other changes to the overall dryer architecture. However, the final drying temperature and target moisture content for tur should be established using appropriate crop-specific post-harvest references and subsequently verified experimentally rather than simply

assuming that the soybean parameters apply to the tur. First, it allows the system to be use for different agricultural commodities. Second, it allows the effect of crop characteristics on drying performance to investigate.

Third, it increases the practical usefulness of the prototype because farmers may handle more than one crop. The comparison will not mean that soybean and tur are directly mixed or dried. Instead, each crop will represent a separate experimental condition. Characteristics of a crop directly influence the design and performance of a drying system. Parameters such as grain size, shape, bulk density, surface area, layer thickness and moisture at which moisture can migrate from the interior of the grain towards the surface.

For each crop, the following parameters should be recorded:

- Initial crop mass
- Initial moisture condition
- Drying-air temperature
- Relative humidity
- Crop mass during drying
- Drying duration
- Energy consumed
- Final moisture condition
- Physical appearance
- Drying uniformity

4.3 Physical Characteristics of Selected Crops:

The physical characteristics of a crop directly influence the design and performance of a drying system for all. Parameters such as grain size, shape, surface area, layer and thickness and moisture distribution affect the resistance that offered to airflow and the rate at which moisture can migrate from the interior of the grain toward the surface of grain. In a forced-convection dryer, heated air must be passed through or around the crop material. If the crop layer is excessively thick, the air may preferentially travel through low-resistance paths rather than the uniformly contacting the entire crop bed. This can result in non-uniform drying process. The proposed dryer therefore requires an arrangement that

provides sufficient exposure of the crop to moving the air while preventing the excessive accumulation of that crop material in a single location.

Soybean is generally handled as a discrete crop material. A grain has a relatively compact structure, and moisture content within the grain must migrate towards the surface before it can be removed by the drying air. The high oil and protein content of soybean makes temperature management particularly important. Excessive drying temperature can contribute to cracking, structural damage, colour changes and other quality loss, while excessive moisture increases the possibility of fungal and other storage-related problems. Therefore, soybean drying in the proposed system must balance two competing requirements:

1. **Sufficient thermal energy must be supplied for timely moisture removal.**
2. **The drying temperature must not be very higher or uncontrolled.**

This is one of the reasons why the proposed IoT control system is important. Rather than allowing the heater to operate continuously, the ESP32 can use measured temperature information to control heater according to a defined set point and control strategy.

Tur is also handled as a grain material, but its physical characteristics differs from soybean. The grain geometry, size distribution, surface properties and internal moisture behaviour leads airflow and drying rate to change. For the selected system, tur should be distributed relatively uniformly across the drying trays. Large or highly uneven layers should be avoided because they can produce differences in drying rate between different regions of the tray. The experimental procedure should maintain consistent loading conditions when comparing different tests. For example, if the dryer is evaluated at different temperatures, the quantity of tur placed inside the dryer should be kept constant wherever possible.

4.4 Initial Moisture Content:

Initial moisture content is one of the most important point in crop drying. The amount of water present in the crop at the beginning of the process describes the quantity of moisture that must be removed before safe storage condition. A crop with high initial moisture requires more drying energy and generally a longer drying period than a crop with lower initial moisture content, assuming that other operating conditions will remain approximately constant. Initial moisture content also affects the weight-loss measurements from initial weight. Since the proposed system includes a load cell, the change in crop mass can be monitored continuously as changes occurs. The decrease in mass during

drying is primarily associated with removal of water or moisture, but the small additional changes may create from handling or other effects.

The wet-basis moisture content is given by:

Mwb = (mw/m)×100

- **Mwb** = moisture content on wet basis (%)
- **mw =** mass of water contained in the sample
- **m** = total mass of wet sample

Since:

m=md+mw

the moisture content can also be expressed as:

Mwb = [(m-md)/m] ×100

Where:

m = initial wet mass

md = dry matter mass.

This equation is important for the proposed system because the load cell measures total crop mass rather than directly measuring water content or moisture directly.

Soybean is harvested or handled over a range of moisture levels, and artificial drying becomes important when the grain moisture is above the targeted storage level. The purpose of the dryer system is not simply to reduce the crop weight. The primary objective is to reduce the crop moisture content to a level suitable for safe storage of crop. Excessive moisture during storage increases the possibility of growth of fungus or unwanted biological growth, heating, insect growth and other mechanisms. The Indian Grain Storage Management and Research Institute notes that high grain moisture promotes conditions favourable to microorganisms and insects and identifies safe moisture as a critical factor in storage management.

For soybean specifically, drying temperature and final moisture level influence the quality during storage. Higher drying temperatures combined with higher final moisture result in greater deterioration than milder drying conditions with lower final moisture. One may have started with relatively low moisture while another may have started with substantially higher moisture. Their final

moisture levels may be different even though the drying duration is identical. One of the major principles of the proposed smart dryer is that drying time alone should not be considered a sufficient indicator of drying completion.

- initial mass,
- mass versus time,
- temperature versus time,
- relative humidity versus time,
- drying duration,
- final mass,
- and, where possible, final moisture content.

This creates a more scientifically defensible assessment of drying performance.

4.5 Recommended Drying Temperature:

Temperature is one of the most important parameter in a crop drying system because it directly affects the rate of moisture removal, energy consumption, drying duration, and final crop quality. In the proposed smart crop dryer, the temperature is not treated as a heating parameter; rather, it is considered a controlled process variable that must be monitored continuously. The proposed dryer is designed to operate in approximately the 45–55°C range. This range is selected as a prototype operating range for evaluating the performance of the dryer and its effect will be evaluated experimentally for both soybean and tur. The two crops have different physical and biochemical characteristics, and that's why their drying behaviour can differ completely. The temperature inside the drying chamber is controlled using the combination of a 24 V PTC heater, 24 V DC blower, ESP32 controller and temperature sensors. while the DS18B20 can provide an additional temperature measurement for comparison and control. drying time alone is not sufficient to determine drying completion.

The smart dryer provides an important advantage over conventional drying process because drying conditions can be recorded continuously throughout the time. The ESP32 can send temperature, humidity and weight-related data, where the data can be stored and visualized or Analyse. Temperature has a direct relationship with crop quality. A higher temperature can increase the rate of evaporation, but excessive temperature can cause thermal damage. For soybean in particular, temperature selection requires care because soybean contains significant amounts of oil and protein. Drying conditions that

are acceptable for one crop may not necessarily be ideal for soybean that's why, the proposed 45–55°C range should be experimentally evaluated rather than assumed to be the optimum condition. Drying occurs when the moisture vapour pressure at the crop surface is higher than the vapour pressure of the surrounding drying air near crop. Hot air alone does not guarantee for the rapid drying. The air must also have sufficient capacity to absorb the additional moisture. Low RH generally improves the moisture-removal capability of the air. A very aggressive drying conditions may produce excessive surface drying while moisture remains present internally within the grain. Airflow is must in a forced-convection dryer because it transfers the heat to the crop and carries evaporated moisture away from the drying chamber of system. The reference drawing specifies approximately 120 m^3/h airflow through the tray stack, with tray open area of at least 40%.

The chamber should be designed so that air is distributed across the complete tray area equally. Air-distribution section below the trays can help to reduce the localized high-velocity flow of air. Soybean and tur are treated separately. They should not be mixed. The proposed system uses approximately five trays, subject to the final mechanical construction. The development of a smart crop dryer provides an alternative approach in which the drying environment can be controlled more effectively. The system combines solar energy with electrical backup, allowing the drying process to continue even when solar radiation is insufficient. The use of a heating element and forced-air blower provides proper control over heat transfer to the crop, while temperature, relative humidity and crop weight are monitored electronically. The incorporation of an ESP32 controller, and IoT technologies further improve the capability of the dryer by allowing continuous monitoring and recording of the drying process. Initially, the harvested crop is weighed and placed on the drying trays. The system is then switched on and the blower establishes air circulation inside the chamber. Electrical energy is supplied to the 24V PTC heater, which increases the temperature of the circulating air. The heated air passes through the crop and transfers heat to the grain surface. As the crop temperature increases, moisture starts to move from the interior, towards the surface and evaporates into the surrounding air. The moving air carries the water vapour away from the crop. The SHT31 continuously monitors temperature and relative humidity, temperature measurement. The load cell detects changes in crop mass while drying progresses. Another important advantage is data availability. In conventional sun drying, the farmer observe the crop visually but does not normally obtain continuous numerical records of temperature, humidity and mass. The proposed IoT dryer can record these parameters throughout the complete drying cycle. The ESP32 can implement a simple hysteresis control method. A set point is defined, and upper and lower limits are established around that set point. When the temperature falls below the lower limit, the heater is activated. When the temperature reaches the

upper limit, the heater is deactivated. This method avoids rapid switching of the heater around a single temperature value.

The crop and drying requirement analysis establishes the scientific foundation for the proposed smart IoT-based crop dryer. The analysis demonstrates that successful crop drying cannot be achieved by supplying heat. The process depends on the interaction between temperature, relative humidity, airflow, crop moisture, crop loading, heat transfer and moisture transfer. Soybean and tur have been selected crops for the experimental study, and their separate processing allows their individual drying characteristics to be investigated. The nominal batch capacity of approximately 5 kg makes the system suitable for a prototype and experimental demonstration. The proposed 45–55°C operating range provides an initial test condition, but its suitability must be confirmed experimentally, particularly with respect to soybean quality. The use of temperature and humidity sensors provides continuous information about the drying environment, while the load cell provides valuable information regarding mass reduction.

CHAPTER – V

MECHANICAL DESIGN OF THE SMART CROP DRYER

CHAPTER - V

MECHANICAL DESIGN OF THE SMART CROP DRYER

5.1 Introduction:

The mechanical design of a smart crop dryer is an important point in developing an efficient and reliable agricultural drying system. The primary purpose of the dryer is to remove moisture from freshly harvested agricultural products under controlled heat or thermal and airflow conditions while maintaining the quality of the product to prevent harm. Conventional open-sun drying is highly dependent on weather conditions and may result in improper drying causing the unwanted bacterial growth, unwanted chemical formation and loss of crops due to fungal growth, due to presence of moisture after the post-harvest losses of crops. A mechanically designed *enclosed* dryer provides a controlled environment where temperature, airflow, drying area, and heat transfer can be controlled. The mechanical design involved the development and integration of several components, including a drying chamber, trays, air inlet, air outlet, heating chamber, fan or blower arrangement, insulation, supporting structure, and heat distribution system. Each component was designed according to the required drying capacity, operating temperature, airflow rate, material properties, manufacturing method, and safety requirements.

The smart crop dryer combines mechanical components with sensors and an automatic control system. However, the effectiveness of the electronic control system depends strongly on the mechanical design of the project. For example, uneven airflow distribution can cause some portions of the crop to dry faster than other portions, whereas some insulation can suck excessive heat loss and increase energy consumption. Therefore, the mechanical design must provide sufficient structural strength, appropriate thermal resistance, uniform airflow, ease of operation, and ease of maintenance. The design consisted of determining the required capacity, estimating the energy requirement, selecting appropriate operating conditions, designing the drying chamber, determining the airflow requirement, selecting the blower, designing the heating chamber and insulation, and finally developing the CAD model and physical prototype.

The drying temperature and drying time were selected according to the characteristics and moisture content present inside the crop. A suitable airflow pattern was established to ensure uniform distribution of heated air throughout the chamber, thereby preventing overall uneven drying. The size

and arrangement of the drying trays were determined to maximize the available drying area and improve the material to airflow.

5.2 Overall Mechanical Design:

The overall mechanical design of the smart crop dryer consists of a rectangular insulated drying chamber supported by a rigid structural frame. The drying chamber contained five trays on which the agricultural product was placed. Heated air was entered into the chamber through an air inlet and distributed across the trays. Moist air was removed through an air outlet positioned to provide continuous circulation.

A typical system can be divided into the following major mechanical components:

- Drying chamber
- Tray assembly
- Heating chamber
- Air inlet duct
- Air outlet duct
- Fan/blower assembly
- Insulation layer
- Supporting frame
- Access door
- Air distribution mechanism
- Fasteners and hinges
- Sensor mounting provisions

The mechanical arrangement should allow air to travel through the heating section before entering the drying chamber. The heated air then passes through the crop placed on the trays inside the closed chamber. As moisture evaporates from the crop, the relative humidity of the air increases and the crop starts to dry. The moist air is subsequently removed through the outlet portion. The chamber should be sufficiently large to accommodate the required quantity of crop while maintaining equal spacing between the trays. Close tray spacing can restrict airflow and produce non-uniform drying.

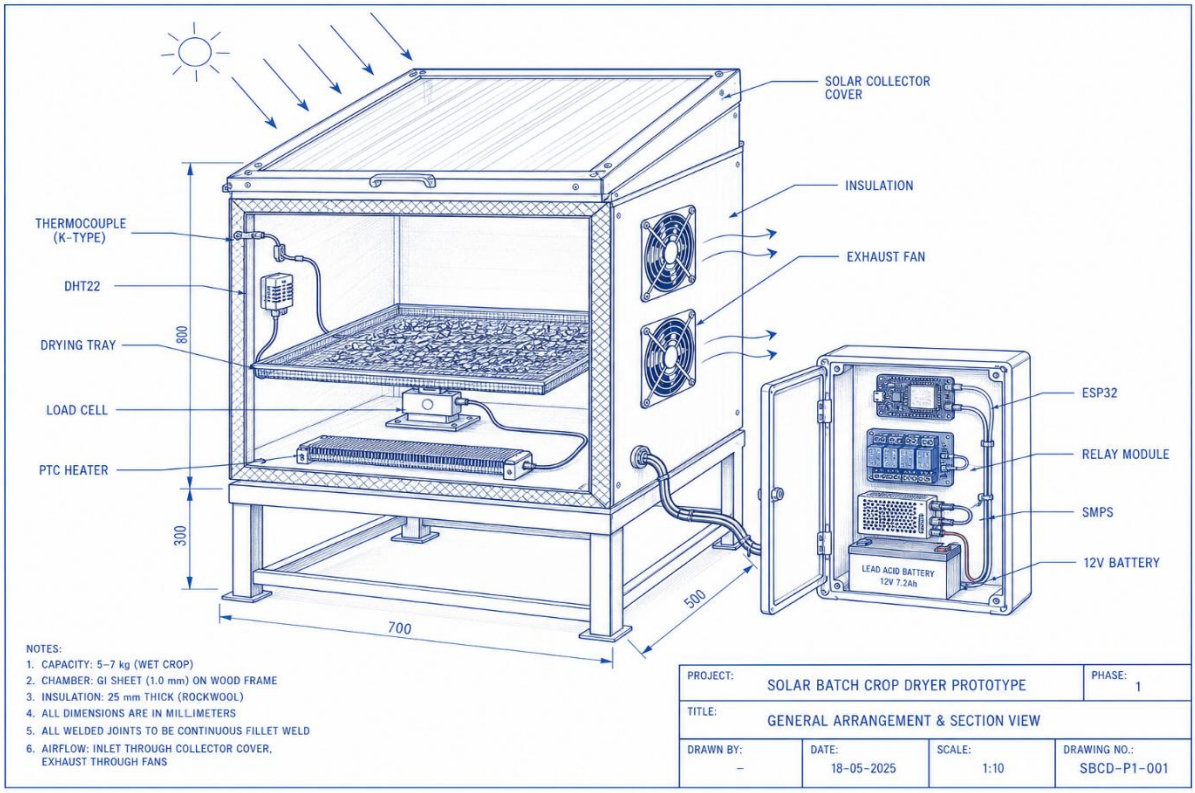


Fig 5.1 General arrangement of the proposed solar batch crop dryer

Ambient air → Air inlet → Fan/blower → Heating chamber → Air distribution section → Drying chamber → Crop trays → Moist-air outlet

5.3 Design of The Drying Chamber:

The drying chamber is the main mechanical component of the crop dryer mentioned in diagram 5.1. It provides an enclosed environment in which moisture is removed from the agricultural product. A rectangular cabinet configuration is arranged in multiple horizontal levels. The internal dimensions were determined from the required tray area and number of trays. Tray spacing is important because it directly affects airflow. If the trays are positioned too close together, the pressure drop increases, and the airflow through the product becomes uneven. The drying chamber should also include an access door. The door may be hinged on one side and provided with a suitable latch or locking mechanism, with dimensions of 24 × 48 inches, while the base and top of it is of 24 × 24 inches dimension.

5.4 Material Selection:

Material selection is an important part of mechanical design because the dryer operates at particular temperatures and comes into contact with agricultural products. The selected material possess:

- Good corrosion resistance
- Adequate mechanical strength
- Thermal resistance
- Easy fabrication
- Easy cleaning
- Low maintenance
- Suitable food-contact characteristics
- Reasonable cost
- Availability in the local market

The drying chamber of the developed Smart Crop Dryer was constructed using a plywood outer layer, thermal sheet insulation, and an aluminium inner layer. This combination provides a practical balance between structural support, thermal insulation, low weight, corrosion resistance, ease of fabrication, and cost.

The wall construction can be represented as:

Plywood outer layer → Thermal sheet insulation → Aluminium inner layer

The outer plywood layer provides mechanical support and protects the insulation from the external environment. The thickness of plywood was of 15mm. Thermal sheet was placed between the two layers as the principal thermal insulation material. The aluminium sheet formed the inner surface of the drying chamber and was exposed to heated air. Plywood was used as the external layer of the drying chamber. It provides a relatively rigid enclosure and helps maintain the overall shape of the dryer. Plywood is also comparatively easy to cut, drill, join, and fabricate.

A thermal sheet was not the first priority for the material, which was glasswool, but it took place of glasswool because of some parameters of the thermal sheet. Thermal sheets are dimensionally stable and robust where glasswool can settle or compress easily.

- Providing an external enclosure for the dryer.
- Supporting and protecting insulation.
- Providing mechanical rigidity to the chamber walls is so important.
- Providing a convenient surface for mounting external components.
- The fabrication cost is lower than that of constructing a complete enclosure from metal sheets.

Because plywood was used near a heated system, its temperature must be kept sufficiently low using thermal sheet insulation. Direct contact between the plywood and heating element should be avoided; therefore, the middle layer of thermal sheet was placed between them.

The thermal sheet was placed between the plywood and aluminum layers. Its primary function was to reduce heat transfer from the drying chamber to the external surroundings. The thermal sheet was suitable for application because it has a relatively low thermal conductivity and good resistance to elevated temperatures. It also helped maintain a more stable temperature inside the drying chamber. Increasing the thickness of the thermal sheet increases the thermal resistance and consequently reduces conductive heat loss. The Thermal sheet was enclosed between the plywood and aluminium layers to protect it from direct mechanical damage and contact with the crop.

An aluminum sheet was used as the inner lining of the drying chamber. Aluminum was selected because it is lightweight, corrosion-resistant, and relatively easy to fabricate. The aluminum layer provides a smooth internal surface that can be cleaned after drying operations. It also protects the thermal sheet from exposure to the drying chamber and provides a durable internal enclosure. The gaps in the aluminum chamber were filled with the silicon sealant after welding to prevent heat leakage.

The functions of the aluminum (inner layer) are as follows:

- The internal wall of the drying chamber was provided.
- Protecting the insulation from the chamber environment.
- This provides smooth and cleanable surface.
- Providing corrosion resistance.
- Withstanding the operating temperature of the dryer.
- Facilitating the fabrication of the internal chamber.

Insulation design is particularly important for the smart crop dryer because the drying process requires the chamber temperature to be maintained above the surrounding ambient temperature. Without sufficient insulation, a significant portion of the supplied thermal energy can be transferred to the surroundings in the chamber. In the developed dryer, a thermal sheet was used as the insulation material and was sandwiched between an external plywood layer and an internal aluminum layer.

The wall arrangement is:

Plywood → Thermal sheet insulation→ Aluminum

The aluminum layer formed the internal surface, while the plywood formed external surface. Thermal sheet occupied the space between these two layers. Particular attention should be given to the heating chamber because it contained the heat source. The thermal sheet should be appropriately positioned to limit the heat transfer to the plywood enclosure. The heating element should not be in direct contact with plywood or other combustible materials because it could harm the model. A suitable metal barrier or separate metal heating enclosure should be provided between the heater and insulation structure. The structural design of the Smart Crop Dryer provides support for the plywood enclosure, thermal sheet insulation, aluminum inner lining, trays, crop load, blower, heating chamber, ducts, and other components. The structural frame was designed in a way that the total weight of the dryer and the crop during operation could show exactly.

The thermal design of the Smart Crop Dryer is based on the requirement to supply sufficient energy to heat the drying air and crop and to remove the moisture contained in the agricultural product. For the developed Smart Crop Dryer, heat loss occurs mainly through the chamber walls, door, bottom, top, and unwanted air leakage. Because the chamber uses a multilayer wall, heat conduction can be analyzed through it. The supporting frame can be fabricated from mild steel sections or from other suitable structural materials. The frame supports the sandwich wall construction and provides mounting points for the fan, heating chamber, trays, and other components. The final prototype of the Smart Crop Dryer consisted of an insulated drying chamber fabricated according to the developed mechanical design. The chamber used a plywood outer layer, thermal sheet insulation, and an aluminum inner lining. The plywood provided the external enclosure and mechanical support, whereas the thermal sheet provided thermal insulation. The aluminum inner layer provided the internal drying surface and protected the insulation from the chamber environment. The trays were positioned inside the aluminum-lined drying chamber at suitable vertical intervals to permit the circulation of heated air. The heating chamber supplied thermal energy to the air, while the fan or blower circulated the air through the system.

Ambient air → Inlet → Blower → Heating chamber → Air distribution → Drying chamber → Trays → Moist-air outlet

The blower supplies the required air circulation. The air was heated in the heating chamber before being introduced into the drying chamber. The heated air passed into the trays chamber where the crop was placed on the trays and removes moisture from the crop. A tray dryer was suitable for small-scale agricultural and food-product drying because the products can be arranged in thin layers on the trays.

CHAPTER – VI

ELECTRICAL AND ELECTRONIC SYSTEM DESIGN

CHAPTER - VI

ELECTRICAL AND ELECTRONIC SYSTEM DESIGN

6.1 Introduction:

The electrical and electronics system of the smart crop dryer forms the fundamental hardware layer through which the electrical power is supplied, distributed, controlled, and monitored in the drying process. The system is designed for an IoT-based smart crop drying application which used for agricultural products such as Tur and Soybean. The electrical design shown in the provided connection diagram combines a 24V DC battery-based power system with solar charging and a 230V AC backup charging source. The generated and stored electrical energy is distributed through a fused, DC distribution arrangement to the major loads and electronic control circuits. The system also incorporates an ESP32 DevKit as the principal microcontroller, temperature and humidity sensing through the SHT31 sensor, temperature measurement through the DS18B20 sensor, and weight measurement using a load cell connected through an HX711 amplifier. The sensing system provides the information required by the control and monitoring system, while the heater and blower constitute the major electrical loads associated with the drying process.

Table 6.1 Electrical and Electronic component specifications

Component	**Specification**
Solar Pannel	25W/ 24V
MPPT Charge Controller	
Lithium Battery	24V
Fuse	25A / 15A / 10A / 5A
DC Distribution Block	100A
Fan / Centrifugal Blower	24V DC
PTC Heater	24V
Buck Converter (24V → 5V)	5A / 3A
AMS!!!&	3.3V

ESP32 DevKit	3.3V(input)
DHT22 Temperature Sensor	
Humidity Sensor	
Moisture Sensor	
Weight Sensor- HX711	
SSR Relay Module 4 Channel	
Adopter	12V 1Amp

The design is based on a 24 V DC electrical architecture. The 24 V battery bank shown in the diagram is rated at 24 V and 18 Ah and acts as the principal stored-energy source for the system. Solar energy is supplied through a 20 A MPPT charge controller to charge the battery bank, while a SMPS connected to a 230 V AC input is shown as a backup charging source. The positive supply path incorporates a 25 A inline ATC/ATC fuse followed by a main DC rocker switch before reaching the 12-way fused DC distribution block. From the distribution block, individual fused branches supply the 24 V PTC heater, 24 V DC blower, and the low-voltage buck converters used for the electronic circuits. This arrangement separates the high-current 24 V loads from the lower-voltage electronic loads and provides individual fuse protection for the outgoing circuits.

The electronic control and sensing portion is centred around the ESP32 DevKit. The diagram shows the ESP32 receiving 5 V DC from an LM2596 buck converter. The ESP32 is connected to the SHT31 temperature and humidity sensor, DS18B20 temperature sensor, and HX711 amplifier. The HX711 provides the interface between the load cell and the ESP32, allowing the electrical signal produced by the load cell to be processed by the microcontroller. The diagram also indicates Wi-Fi communication between the ESP32 and the MQTT-based monitoring architecture. In the shown architecture, the ESP32 sends data through MQTT, while the and its associated Node-RED, Influx DB, and Grafana components are shown as part of the original monitoring system. The electrical and electronic design therefore provides two closely related functions. The first function is the delivery of electrical power to the drying equipment, particularly the heater and blower, while the second function is the acquisition of physical parameters from the drying chamber and the transmission of those measurements through the IoT communication system. The combination of these functions allows the crop drying system to operate as a smart dryer rather than as a conventional manually operated drying unit.

6.2 Electrical System Architecture:

The electrical architecture shown in the connection diagram is organized around a 24 V DC supply system. The solar panel is connected to a 20 A MPPT charge controller, which in turn is connected to the 24 V 18 Ah battery bank. The solar panel provides the primary renewable charging source represented in the diagram. The MPPT charge controller regulates the charging connection between the solar source and the battery bank. The battery therefore acts as the energy storage element of the system and provides the electrical source required when the solar input is insufficient or unavailable. A second charging path is provided through the SMPS identified as the backup charging unit. The SMPS receives its input from a 230 V AC source and provides DC output. The positive output of the SMPS is shown connected into the positive battery-side supply path, while the negative output is connected to the common negative return. A diode is shown in the positive connection between the charging source and the battery-side circuit. The diagram therefore represents a system in which solar charging and AC backup charging can support the 24 V battery system.

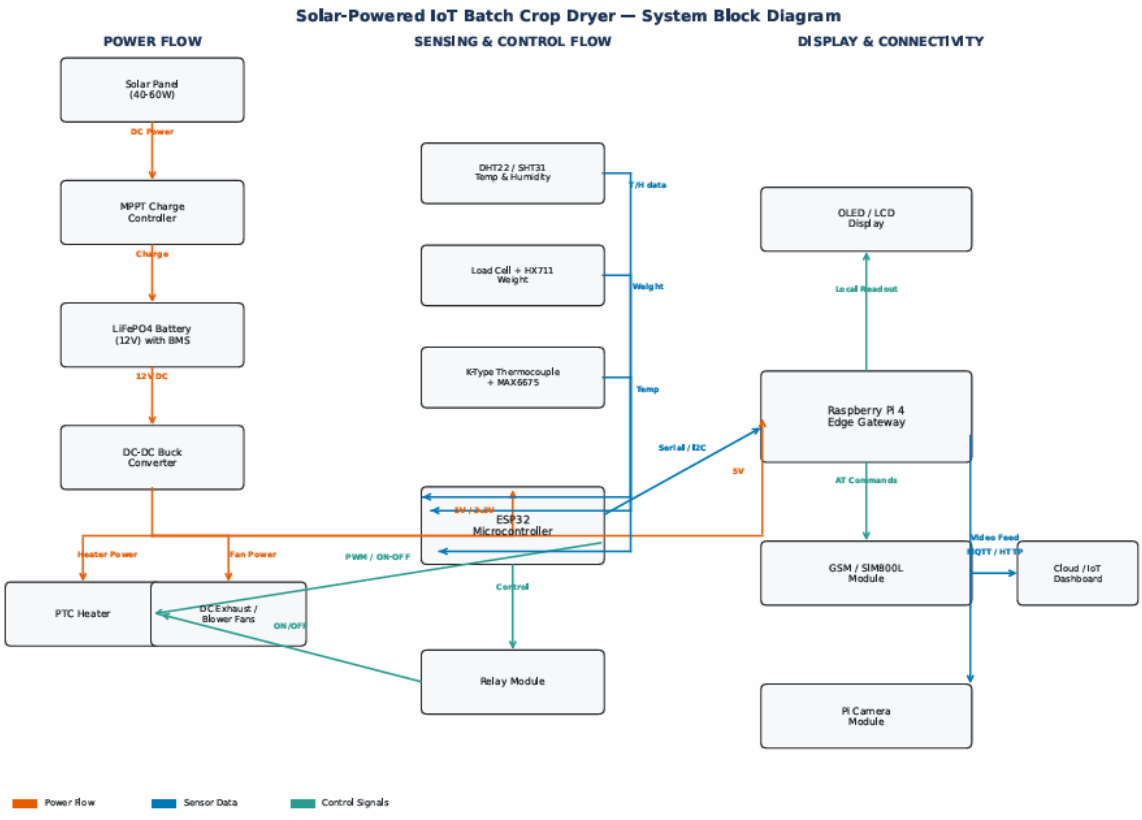


fig 6.1 System block diagram of the IOT Solar Crop Dryer

The positive battery output is routed through a 25 A inline ATC/ATC fuse as per mentioned in diagram 6.1. The fuse is located before the main DC rocker switch and therefore forms the principal protective element on the main positive supply path. After the fuse, the positive supply passes through the main DC rocker switch. The switched 24 V line is then connected to the input of the 12-way fused DC distribution block. The distribution block provides separate outgoing circuits for the various loads represented in the diagram. The negative battery terminal is connected to the common negative return, and the diagram identifies the ground or zero-volt conductors as a common connection throughout the electrical system. The 12-way fused distribution block is rated at 100 A according to the diagram. It provides the central point for distributing the 24 V DC supply to the drying loads and the low-voltage electronics. The heater branch is protected by a 25 A fuse, while the blower branch is protected by a 15 A fuse. Separate fused branches are shown for the buck converters supplying the electronic control circuits. A 10 A fuse is shown on the branch feeding the 24 V-to-5 V, 5A buck converter, while a 5 A fuse is shown on the branch associated with the 24 V-to-5 V, 3 A LM2596 buck converter.

This arrangement creates a clear separation between the main energy-storage circuit and the individual load circuits. The heater and blower remain connected to the 24 V system, while the ESP32 and sensor electronics are supplied through regulated 5 V conversion stages. The architecture consequently allows the 24 V battery system to directly support the high-power components while avoiding the direct application of 24 V to the low-voltage electronic devices. The electrical architecture is particularly relevant to a crop dryer because the heater and blower require substantially different electrical characteristics from the sensing and communication electronics. The heater and blower are shown as 24 V DC loads and are therefore connected to the fused 24 V distribution system. In contrast, the ESP32 DevKit and associated electronic components require a lower-voltage supply, which is obtained using buck conversion from the 24 V system.

6.3 Microcontroller Selection:

The ESP32 DevKit is the microcontroller platform shown in the supplied electrical and electronic connection diagram. It is used as the central controller for the sensor interface and IoT communication portion of the smart crop dryer. The diagram places the ESP32 between the physical sensing components and the MQTT communication path. The sensor connections shown include the SHT31 temperature and humidity sensor, the DS18B20 temperature sensor, and the HX711 load-cell amplifier. The ESP32 is supplied from the 5 V output of the LM2596 buck converter. The LM2596 converter receives the 24 V DC system supply and converts it to a lower regulated voltage suitable for the controller. The diagram identifies the controller-side supply as 5 V and shows a common

ground connection between the buck converter and ESP32. This arrangement ensures that the controller is electrically separated from the direct 24 V supply while remaining powered from the same battery-based electrical system.

The ESP32 performs the electronic interface function between the sensors and the IoT monitoring system. The SHT31 sensor is connected through the I2C interface, with SDA and SCL lines represented separately in the diagram. The DS18B20 provides a temperature data connection to the ESP32, while the HX711 amplifier provides the interface required for the load cell. The resulting measurements can therefore be acquired by the ESP32 and transferred through its Wi-Fi capability using MQTT. The use of the ESP32 is therefore directly reflected in the supplied system architecture rather than being an independently introduced component. The controller is consequently an important part of the smart crop dryer because it connects the physical drying process with the software monitoring system. Measurements from the drying environment and crop-related parameters can be acquired at the controller and communicated through MQTT. This provides the electronic foundation for implementing IoT-based monitoring of the drying operation.

6.4 Power Supply Design:

The power supply design of the smart crop dryer is based on a 24 V, 18 Ah battery bank. The battery bank is the central energy-storage element shown in the electrical connection diagram. The battery is connected to the solar charging system through the 20 A MPPT charge controller. The controller regulates the electrical energy supplied by the solar panel to the battery. The system also incorporates a heater connected to a 230 V AC input as a backup charging source. The solar panel is connected to the positive and negative input terminals of the 20 A MPPT charge controller. The battery terminals of the charge controller are then connected to the 24 V battery bank. The diagram labels the positive and negative battery connections as BAT+ and BAT. This arrangement establishes the solar charging path between the photovoltaic source, charge controller, and battery.

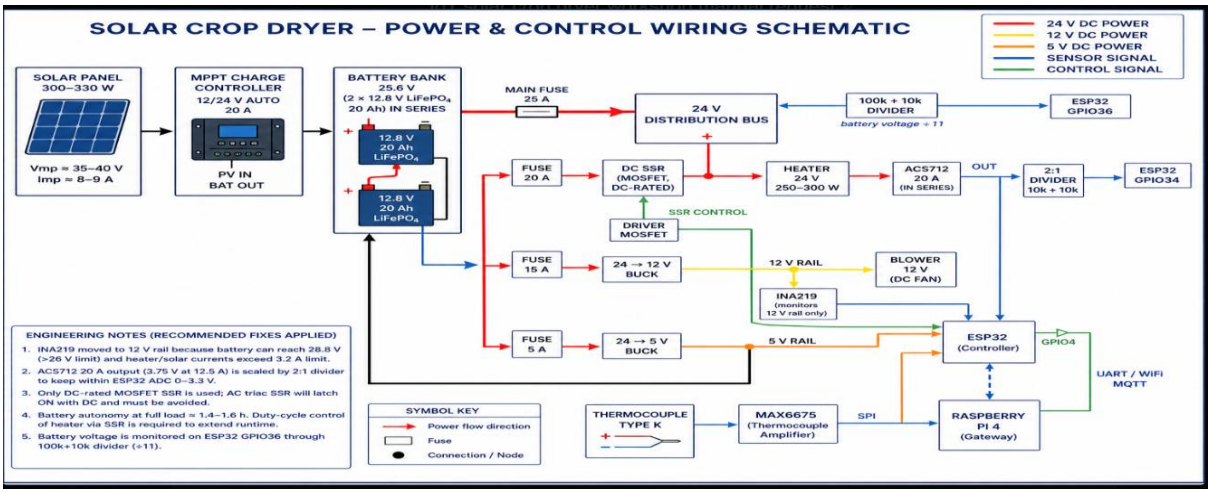


Fig 6.2. Power and Control wiring schematic

The backup charging path begins at the 230 V AC input. The live and neutral conductors of the AC input are connected to the input of the DC SMPS. The SMPS provides a 24 V DC output. Its positive output is connected into the positive supply path leading toward the battery, with a diode represented in that path. Its negative output is connected to the common negative return. The presence of this second supply path provides an alternative means of supporting the battery system when solar energy is not adequate. After the battery and charging arrangement, the 24 V supply enters the load-distribution section. The main positive conductor passes through the 25 A inline ATC/ATC fuse and the main DC rocker switch before reaching the fused distribution block. The distribution block then provides individually fused outputs to the heater, blower, and DC-DC converter circuits. The electronic circuits operate from regulated 5 V supplies generated from the 24 V battery system. The diagram shows a 24 V-to-5 V, buck converter associated with the ESP32.

6.5 Heating Element:

The heating element shown in the diagram is a 24 V PTC heater. It is connected directly to one of the fused outputs of the 12-way DC distribution block. The heater branch is protected by a 25 A fuse, as indicated in the connection diagram. The positive output from the distribution block passes through the 25 A fuse and then connects to the 24 V PTC heater. The negative side of the heater is returned to the common negative bus. The PTC heater is therefore operated from the 24 V DC battery-based supply. Its position after the main fuse, main DC rocker switch, and distribution block means that the heater receives power only after the main electrical system has been energized and the corresponding distribution branch is supplied. The branch fuse provides dedicated protection for the heater circuit.

- Voltage
- Wattage
- Current
- If wattage is known:
- Current = Power / Voltage

Within the smart crop dryer, the heater represents the primary electrical source responsible for supplying thermal energy to the drying system. The electrical diagram does not specify the exact wattage or physical construction of the PTC heater beyond its 24 V designation, and therefore no additional numerical heating characteristics are assumed here. The heater is electrically integrated with the 24 V distribution architecture rather than the 5 V electronic circuit. This distinction is important because the controller and sensors operate through the low-voltage regulated supply, while the heater remains a 24 V load. The electrical separation reduces the possibility of directly applying the high system voltage to the sensor and controller circuitry.

For the intended drying of tur, soybean the heater forms part of the controlled drying environment. The environmental temperature measured by the temperature sensors can be used by the control program to monitor the drying condition. However, the supplied diagram does not show a dedicated heater relay, solid-state switching device, or detailed heater-control driver.

6.6 Fan/Blower System:

The air circulation component shown in the connection diagram is a 24 V DC blower. The blower is connected to the 12-way fused DC distribution block through a dedicated 15 A fuse. Its positive conductor originates from the fused distribution section and is routed through the 15 A protection element to the blower. The negative conductor is connected to the common negative return. The blower is an important electrical load in the crop dryer because the drying system requires air movement through the drying chamber. In the diagram, the blower operates from the same 24 V DC system supplied by the battery bank. Its electrical branch is kept separate from the heater branch, and the 15 A fuse provides individual branch protection.

The diagram identifies the blower specifically as a 24 V DC blower and does not provide a separate motor-driver circuit. The electrical architecture confirms the blower's supply voltage and fuse protection, but it does not establish a particular control device. The blower and heater together represent the main drying loads. The heater provides the thermal component of drying, while the

blower provides air movement. The electronic sensors allow the environmental conditions to be monitored by the ESP32. This relationship enables the smart crop dryer to associate the electrical operation of the drying equipment with measured temperature, humidity, and crop weight information.

6.7 Temperature Sensor:

Temperature measurement in the proposed system is represented by the DS18B20 temperature sensor and the SHT31 sensor. The DS18B20 is shown as a separate temperature sensor with VCC, DQ, and GND connections. A 4.7 kΩ pull-up resistor is shown between the data line and the positive supply. The data connection is routed to the ESP32, while the power and ground connections are connected to the corresponding controller supply rails. The DS18B20 therefore provides a dedicated digital temperature measurement path to the ESP32. The 4.7 kΩ pull-up shown in the diagram is an important part of the sensor interface because it connects the data line to the supply side of the circuit. The data line then carries the sensor information to the microcontroller. The diagram also shows the SHT31 temperature and humidity sensor. The SHT31 module contains VCC, SDA, SCL, and GND connections. The SDA and SCL lines are connected to the ESP32 as the I2C communication interface. This means that temperature information is available from both the SHT31 and the DS18B20, while the SHT31 additionally provides humidity information.

The temperature measurements are particularly significant for the crop dryer because temperature is one of the main environmental parameters associated with drying. For tur, soybean, the temperature condition of the drying environment can be monitored continuously through the electronic sensing system.

6.8 Humidity Sensor:

Humidity measurement is provided by the SHT31 sensor shown in the lower section of the supplied diagram. The sensor is labelled as an SHT31 temperature and humidity sensor and contains four electrical connections identified as VCC, SDA, SCL, and GND. The sensor communicates with the ESP32 through the I2C interface.

The VCC connection provides power to the sensor, while GND is connected to the common ground of the electronic system. SDA and SCL form the two communication lines between the SHT31 and the ESP32. The diagram clearly identifies SDA and SCL as I2C signals, and the legend further identifies the corresponding electrical lines as I2C data and I2C clock. Humidity monitoring is an important function of the smart crop dryer because the moisture present in the drying environment changes as the crop releases water during drying. The SHT31 allows the environmental humidity to

be measured and transferred to the ESP32. The microcontroller can then incorporate the measured value into the IoT monitoring data transmitted through MQTT.

The supplied design does not specify a separate humidity-control circuit. The role of the SHT31 shown in the diagram is therefore treated as measurement and monitoring. The measured humidity value can form part of the drying-process data together with temperature.

6.9 Moisture Monitoring Approach:

A separate crop moisture sensor is not visibly represented in the supplied connection diagram. The diagram identifies temperature measurement through the DS18B20, temperature and humidity measurement through the SHT31, and weight measurement through the load cell and HX711 amplifier. It does not show a separate sensor labelled specifically as a crop moisture sensor, nor does it provide its power, signal, interface, or connection to the ESP32.

6.10 Weight Sensor/Load Cell:

The crop weight measurement system consists of a load cell and an HX711 amplifier module. The load cell is shown in the lower-left section of the electronic circuit diagram. Four conductors are represented between the load cell and HX711 module, corresponding to E+, E−, A+, and A−. These connections form the electrical interface between the strain-gauge-based load cell and the HX711 measurement circuit. The HX711 is identified as the load-cell amplifier in the diagram. It provides VCC, DT, SCK, and GND connections on its controller side. The load cell is connected to the E+, E−, A+, and A− terminals of the HX711. The HX711 is then connected to the ESP32 using its DT and SCK signals, together with the supply and ground connections.

- **Load Cell → HX711 → ESP32**

The load cell converts the mechanical load associated with the crop into an electrical signal. The HX711 provides the amplification and digital interface required for the ESP32 to obtain the measured weight. This arrangement makes the load-cell subsystem an important part of the smart crop dryer because crop weight can be monitored during the drying process. As drying proceeds, the mass of the crop can change because moisture is removed from the product. Consequently, the weight measurement provides information that complements the environmental measurements obtained from the temperature and humidity sensors. The combination of crop weight and environmental conditions provides a broader representation of the drying process. The diagram does not provide the load-cell capacity, calibration factor, accuracy, or mechanical mounting arrangement. The electrical design itself establishes the load-cell-to-HX711 and HX711-to-ESP32 connections.

6.11 Relay and Switching Circuit:

The supplied electrical diagram clearly shows a main DC rocker switch in the 24 V power path. The switch is located after the 25 A inline ATC/ATC fuse and before the 12-way fused DC distribution block. This switch therefore functions as the main manual switching element for the DC system. When the main switch is turned on, the protected 24 V supply can reach the distribution block and its connected branches. A the circuit contains a main DC rocker switch.

6.12 Motor Driver:

The supplied connection diagram identifies the 24 V DC blower as a load but does not show a dedicated motor-driver circuit. No specific motor-driver module, transistor is labelled between the 12-way distribution block and the blower. The blower is shown connected through a 15 A fuse to the 24 V DC distribution system. Accordingly, the electrical design establishes the blower's 24 V supply and its 15 A branch protection, but it does not provide enough information to identify a particular motor-driver topology. A motor-driver description would not accurately represent the supplied circuit.

6.13 Sensor Placement:

The sensor placement shown by the electrical diagram is primarily defined through the sensor connections rather than through a detailed mechanical layout of the drying chamber. The SHT31 is shown as the environmental temperature and humidity sensing device, while the DS18B20 is shown as a separate temperature sensor. The load cell is associated with the crop weight measurement system and is connected to the HX711 amplifier.

The SHT31 should therefore be understood as the environmental sensing element of the electronic system, providing both temperature and humidity measurements to the ESP32 through the I2C interface. The DS18B20 provides an additional temperature measurement channel. The load cell is electrically connected to the HX711 and is mechanically associated with the crop load whose weight is being measured.

DS18B20 → digital temperature sensor → ESP32 - SHT31 → temperature + humidity → I2C → ESP32

6.14 Electrical Circuit Diagram:

The supplied electrical circuit diagram represents the complete relationship between the major power components, drying loads, sensors, controller, and IoT communication elements. The upper portion of the diagram is dedicated primarily to the 24 V power system. The solar panel feeds the 20 A MPPT charge controller, which is connected to the 24 V 18 Ah battery bank. A 230 V AC input is also connected to a SMPS identified as the backup charging source. The outputs of the charging arrangement are connected to the battery-side DC system.

From the battery, the principal positive conductor passes through the 25 A inline ATC/ATC fuse and the main DC rocker switch. The switched positive supply then enters the 12-way fused DC distribution block. The distribution block has separate outgoing fused circuits. The diagram shows a 25 A fuse feeding the 24 V PTC heater and a 15 A fuse feeding the 24 V DC blower. Other fused outputs supply the low-voltage buck-converter circuits.

6.15 Power Consumption:

The power-consumption analysis of the smart crop dryer is based on the electrical loads and power-supply arrangement represented in the supplied diagram. The major 24 V loads are the PTC heater and DC blower. The electronic system consists of the ESP32, SHT31, DS18B20, and HX711, which are supplied through the low-voltage conversion stage.

- PTC heater
- Blower
- ESP32
- SHT31
- DS18B20
- HX711
- Other actual loads

Use:

- $P = V \times I$
- Energy = Power × Time

Battery energy = 24 × 18 = 432 Wh

6.16 Electrical Safety:

Electrical safety is incorporated into the system through the use of fuses, a main DC switch, separate load branches, and a common return arrangement. The main positive battery supply contains a 25 A inline ATC/ATC fuse before the main DC rocker switch. This fuse provides primary protection for the downstream main DC supply path. The main rocker switch provides a means of manually disconnecting the switched 24 V distribution system. The 12-way DC distribution block provides individual fused circuits for the connected loads. The PTC heater is shown with a 25 A fuse, and the blower is shown with a 15 A fuse. Additional fuse protection is shown on the buck-converter branches, including the 10 A and 5 A protection elements indicated in the diagram. This individual branch protection helps prevent a fault in one downstream circuit from directly affecting every other branch.

Correct polarity is also an important safety requirement in the system. The diagram distinguishes positive and negative conductors, and its notes specifically emphasize observing correct polarity throughout the system. The common ground arrangement shown in the diagram means that the negative returns form the common 0 V reference for the relevant DC circuits. The 230 V AC input to the backup SMPS requires particular care because it is the only explicitly shown mains-voltage portion of the system. The AC input is identified as 230 V AC and is connected to the SMPS input terminals. The AC portion should therefore be physically isolated and suitably enclosed in the final hardware.

6.17 Hardware Integration:

Hardware integration is the stage at which the individual electrical, electronic, sensing, and power components represented in the connection diagram are combined into one functional smart crop dryer system. The integration begins with the 24 V power source. The solar panel and MPPT charge controller form the renewable charging section, while the 230 V AC SMPS provides the backup charging path. Both support the 24 V battery-based electrical system shown in the diagram. The battery output is connected to the main protected supply through the 25 A inline fuse and main DC rocker switch. The switched supply reaches the 12-way fused DC distribution block, from which the principal drying loads and low-voltage electronics are supplied. The heater receives 24 V through its 25 A fused branch, while the blower receives 24 V through its 15 A fused branch. The electronic control system is integrated through the 24 V-to-5 V conversion stage. The LM2596 buck converter supplies 5 V to the ESP32 DevKit. The SHT31 is connected to the ESP32 through its I2C interface, and the DS18B20 is connected through its digital data connection with the 4.7 kΩ pull-up shown in

the diagram. The load cell is connected to the HX711 amplifier, which then communicates with the ESP32 through DT and SCK.

The completed hardware therefore consists of an interconnected power subsystem, drying-load subsystem, sensing subsystem, controller subsystem, and IoT communication subsystem. The power subsystem provides the required 24 V and regulated 5 V supplies. The drying-load subsystem consists of the 24 V PTC heater and 24 V DC blower. The sensing subsystem consists of the SHT31, DS18B20, and load cell with HX711 amplifier. The ESP32 provides the processing and wireless communication interface. Together, these components form the electrical and electronic foundation of the IoT-based smart crop dryer intended for tur, soybean. The integration of these components is significant because the system does not treat drying as only a heating operation. The electrical hardware enables the environmental conditions and crop weight to be monitored while the drying equipment operates from the 24 V electrical system. The resulting combination of power control, sensing, embedded processing, and IoT communication establishes the hardware platform required for smart monitoring of the crop-drying process.

CHAPTER – VII

ELECTRICAL AND ELECTRONIC SYSTEM DESIGN

CHAPTER - VII

ELECTRICAL AND ELECTRONIC SYSTEM DESIGN

7.1 Introduction

The IoT has become a very important technology for the intelligent framework for developing agricultural systems. which is helpful for developing intelligent agricultural systems. IoT enables devices such as sensors, microcontrollers, actuators, and communication modules to collect, process, exchange, and monitor data through a network. In agricultural applications, IoT can be used to automate environmental monitoring, crop drying, irrigation, greenhouse management, storage, and other operations. Which makes control decisions according to predefined operating The IoT- Based Smart Crop Dryer uses agricultural technology to monitor and control the drying environment automatically based on the crop's drying capacity. The basic and initial components and primary parameters considered in the system are temperature, relative humidity, and drying conditions. Sensors continuously measure the environmental conditions. Inside the drying chamber. The sensor data are processed by a microcontroller. It controls the devices such as the heater and ventilation fan automatically. When the temperature inside the dryer is lower than the required level, the heating system can be activated. When the temperature becomes sufficiently high, the heater can be switched OFF. Similarly, the ventilation fan can be controlled to remove moist air from the crop drying chamber.

The IoT communication system provides an additional advantage because the measured parameters can be transmitted to a remote monitoring platform. This allows the user to observe the operating condition of the crop dryer without continuously standing near the equipment. The architecture therefore combines sensing, processing, Communication, control, and monitoring into a single integrated system.

7.2 Objectives of the IoT Architecture

The main objectives of the proposed IoT architecture are:

- To continuously monitor the environmental conditions

Inside the crop- drying chamber.

- To acquire temperature and humidity data using sensors.

- To automatically control the heater and fan.
- To process sensor data using a microcontroller.
- To transmit sensor information through an IoT

Communication network.

- To provide remote monitoring of the drying process.
- To reduce the requirement for continuous manual

Supervision.

- To improve consistency in the drying process.
- To prevent excessive temperature that could damage the

Crop.

- To provide a scalable architecture that can be modified for different crops.

7.3 Overall System Architecture

The proposed smart crop dryer consists of several interconnected functional layers. These layers work together to perform sensing, decision-making, actuation, communication, and monitoring.

The major components of the architecture are:

1. Crop Drying Chamber
2. Temperature and Humidity sensors
3. Microcontroller
4. Heater
5. Exhaust/ventilation Fan
6. Relay or switching circuit
7. IoT communication Network
8. Cloude/IoT monitoring platform
9. User Interface

10. Power supply

The general data flow can be represented as:

Crop → Sensors → Microcontroller → Control Algorithm →Actuators

At the same time:

Sensors → Microcontroller → Internet → IoT Platform → User

7.4 Layered IoT Architecture

The proposed system can be represented using a five- layer IoT architecture.

7.4.1 physical /sensing Layer

The sensing layer consists of sensors installed inside or around the drying chamber. The main purpose of this to collect information about the drying environment.

Typical measurements include:

- Temperature
- Relative humidity
- Crop moisture content, where a suitable moisture sensor is available
- Air conditions

The sensor outputs are supplied to the microcontroller.

7.4.2 Processing layer

The processing layer consists primarily of the microcontroller

It receives raw sensor information and processes it into useful data.

The controller performs operations such as:

- Sensor reading
- Data reading
- Data conversion
- Data filtering

- Threshold comparison
- Data transmission
- Actual control
- Communication
- Fault detection

For an ESP32-based implementation, it is suitable because it combines processing capability with built-in wireless connectivity

7.4.3 Communication layer

The communication layer provides connectivity between the smart crop dryer and the remote IoT platform. A Wi-Fi-enabled microcontroller can connect to a wireless network and transmit sensor measurements through the internet.

The communication process can be expressed as:

Sensor → ESP32 → Wi-Fi Router → Internet → IoT Server →

User Interface depending on the selected IoT platform, protocols such as HTTP or MQTT can be used.

7.4.4 Application Layer

The application layer provides information to the user.

The user interface may display:

- Current temperature and humidity
- Humidity-specific temperature
- Heater status
- Fan status
- Drying status
- Sensor readings
- Historical measurements
- Warning or alarm conditions

This layer allows the operator to understand the current

condition of the dryer without directly inspecting the

7.5 Sensor Layer

The sensor is the device and most important component for

The crop dryer, which controls the chemical and biological

Environmental detection is done by the sensor, such as heat,

light, or pressure, which is combined in a safe way. The sensor most important and smartest section of the machine which dries the crop with specific accuracy. The sensor senses the output, which is measurable

7.6 Temperature Sensor

Temperature is most important for the crop dryer; a specific temperature parameter to control with a sensor gives you control of the major part. Quick changes: this device uses raw physical changes with numerical sensing components to improve the quality in the needed way. It is critical for time management; insufficient temperature can increase, while the temperature may not affect the inside crop. It provides the crop with the moisture containing them.

The temperature sensor continuously checks the inside chamber temperature with proper measurement with microcontroller.

The measured temperature can be represented as :

- T = temperature measured inside the drying chamber
- Where T is generally expressed in degrees Celsius (°C)
- The components measure the specific temperature range.

For example :

$$T < T_{min}$$

Including the heating required as heating.

Similarly,

$$T > T_{max}$$

The various heating requirements are stopped, including

The exact T_{min} and T_{max} value can be determined experimentally. This crop beginning will validation bring crop.

The electrical device which measure the heat of objects or environmental and thermal conditions.

7.6.1 Humidity sensor

A humidity sensor measures the humidity parameter from the chamber as relative humidity. During the drying of the crops, water is transferred from the crop drying chamber to the surrounding air. Therefore, all the components of humidity which are in the surrounding air will be removed moist air from the crops.

Relative humidity is represented as:

$$RH = \frac{P_v}{P_s} \times 100$$

Where:

- RH = relative humidity (%)
- P_v = Actual water vapour pressure
- P_v = saturated vapour pressure

Moisture humidity represents the actual vapour inside the chamber. Where all the humidity is being removed with the help of temperature and pressure comparatively drier air.

7.7 Microcontroller Architecture

The microcontroller is the central processing unit that controls memory (RAM and ROM) and handles input/output (I/O) operations with the smart crop dryer.

In the ESP32-based implementation provided:

- Sensor interfacing
- Input /output analog
- Use Wi-Fi communication
- Process the data
- Logical control

- IoT system communication
- Analogue and Digital interfacing
- Connectivity subsystem

We have the current microcontroller cycle:

Temperature Normal

↓

Normal Control

↓

Temperature High

↓

Heater OFF

↓

Fan ON

Temperature Critical

↓

Heater OFF

↓

Fan ON

↓

Alarm

This loop continuously dries the crop with this cycle and changes the conditions.

7.8 Actuator Layer

This makes the actual layer converts electrical signals into physical action

The main actuators in the proposed dryer are:

- The heating element
- Ventilation/exhaust fan
- Optional air circulation fan

7.9 Heating System

The heating element supplies thermal energy to the drying chamber.

When the measured temperature is below the required operating range, the controller can activate the heater through a relay or suitable power-control circuit.

The basic control condition is:

$$\text{If } T < T_{set} - H,\ Heater = ON$$

and

$$\text{If } T > T_{set} + H,\ Heater = OFF$$

where:

- T_{set} = desired temperature
- H = hysteresis value

The use of hysteresis prevents the heater from switching ON and OFF repeatedly around a single temperature value.

7.10 Ventilation Fan

The fan removes humid air from the drying chamber and helps maintain air movement.

The fan can be controlled according to temperature, humidity, or both.

For example:

$$RH > RH_{max} \Rightarrow Fan = ON$$

This causes humid air to leave the chamber.

The fan may also be activated when the temperature becomes excessive:

$$T > T_{max} \Rightarrow Fan = ON$$

This can assist with temperature reduction, depending on the dryer design.

7.11 IoT Communication Architecture

The IoT communication system allows sensor data to be transmitted from the dryer to a remote monitoring platform.

The communication architecture is:

Sensors

↓

ESP32

↓

Wi-Fi

↓

Internet

↓

IoT Platform

↓

Database/Dashboard

↓

Mobile/Computer

↓

User

The ESP32 periodically sends information such as:

- Temperature = 52.4 °C
- Humidity = 38 %
- Heater = ON
- Fan = ON
- Drying Status = ACTIVE

The user can then monitor the dryer remotely.

7.11 Wi-Fi Communication

Wi-Fi provides wireless connectivity between the microcontroller and the local network.

The basic sequence is:

1. The ESP32 starts.
2. The Wi-Fi interface is initialized.
3. The controller searches for the configured network.
4. Authentication takes place.
5. An IP connection is established.
6. Sensor data are transmitted to the IoT platform.
7. The connection is periodically checked.

If the connection is interrupted, the controller can attempt reconnection.

A robust system should distinguish between:

- Sensor failure; Wi-Fi failure
- IoT server failure
- Power failure
- Actuator failure

CHAPTER – VIII

CONTROL ALGORITHM AND AUTOMATION

CHAPTER - VIII

CONTROL ALGORITHM AND AUTOMATION

8.1 Introduction

Automation is one of the most important features of the proposed IoT-Based Smart Crop Dryer. Conventional crop-drying methods generally depend on manual observation and control of environmental conditions. The operator may need to periodically check the temperature, humidity, and condition of the crop and manually operate the heating and ventilation systems. Such an approach can result in inconsistent drying, unnecessary energy consumption, excessive heating, and increased human effort. The proposed smart crop dryer overcomes these limitations by incorporating an automatic control system. Sensors continuously measure the environmental conditions inside the drying chamber, while a microcontroller processes the sensor data and determines the appropriate operation of the heater and fan. The control algorithm is based primarily on temperature and relative humidity thresholds. The measured parameters are compared with predefined reference values. According to these comparisons, the system automatically turns the heating and ventilation devices ON or OFF. The IoT component further improves the system by allowing the operating parameters to be monitored remotely. Thus, the system combines automatic local control, safety control, and IoT-based monitoring.

The basic control concept is:

Sensors → Microcontroller → Control Algorithm → Actuators

where the actuators mainly consist of the heating element and ventilation fan.

8.2 Objectives of the Control Algorithm

The main objectives of the control algorithm are:

- To maintain the required drying temperature.
- To monitor humidity continuously.
- To automatically operate the heater.
- To automatically operate the ventilation fan.
- To prevent excessive temperature.

- To reduce unnecessary energy consumption.
- To maintain suitable drying conditions.
- To reduce manual intervention.
- To provide safe operation of the dryer.
- To transmit the operating status to the IoT platform.
- To detect abnormal operating conditions.
- To provide reliable and repeatable drying operation.

8.3 Control Variables

The performance of the dryer depends on several parameters. The most important variables considered by the proposed control system are temperature and humidity.

8.3.1 Temperature

Temperature is the primary control variable.

It is represented by:

$$T = \text{Measured chamber temperature}$$

and is expressed in °C.

The controller compares the measured temperature T with the desired temperature T_{set}.

For example:

$$T < T_{set}$$

indicates that additional heating may be required.

If:

$$T > T_{set}$$

the controller may reduce or stop heating depending on the selected control strategy.

8.3.2 Relative Humidity

Relative humidity is represented by:

$$RH = \text{Measured relative humidity}$$

and is expressed as a percentage.

During drying, moisture is transferred from the crop to the surrounding air. Therefore, monitoring relative humidity provides useful information about the condition of the drying chamber. A high humidity condition can be used to activate the ventilation fan.

$$RH > RH_{max} \Rightarrow Fan = ON$$

8.3.3 Actuator Status

The control system also maintains the state of each actuator.

For example:

$$H \in \{0, 1\}$$

where:

- $H = 0 \rightarrow$ Heater OFF
- $H = 1 \rightarrow$ Heater ON

Similarly:

$$F \in \{0, 1\}$$

where:

- $F = 0 \rightarrow$ Fan OFF
- $F = 1 \rightarrow$ Fan ON

8.4 Set-Point Selection

The set point represents the desired operating condition of the dryer.

The appropriate temperature depends on:

- Crop type
- Initial moisture content
- Final required moisture content
- Crop quality requirements
- Dryer design
- Drying duration

- Airflow conditions

Therefore, the temperature set point should not be considered universal for every crop.

The system can be configured using:

$$T_{set}$$

and humidity limits:

$$RH_{max}$$

For a particular experimental setup, these values should be determined from the crop-drying requirements and validated experimentally.

8.5 Temperature Control Strategy

The simplest control method used in the proposed system is threshold-based control.

The heater is switched ON when the temperature falls below the lower control limit.

$$T < T_{low} \Rightarrow Heater = ON$$

When the temperature reaches the upper control limit:

$$T > T_{high} \Rightarrow Heater = OFF$$

This creates a temperature operating band:

$$T_{low} \leq T \leq T_{high}$$

The use of two limits instead of a single threshold introduces hysteresis.

8.6 Hysteresis Control

Hysteresis is important because it prevents rapid switching of the heater.

If the heater were switched ON whenever:

$$T < T_{set}$$

and OFF whenever:

$$T > T_{set}$$

small temperature fluctuations around the set point could cause repeated ON/OFF switching.

Instead, the controller uses:

$$T_{low} = T_{set} - H$$

$$T_{high} = T_{set} + H$$

where H is the hysteresis value.

The control logic becomes:

IF Temperature < Tlow

Heater = ON

ELSE IF Temperature > Thigh

Heater = OFF

ELSE

Maintain previous heater state, this produces more stable operation and reduces unnecessary switching.

8.7 Humidity Control

Humidity control is performed using the relative humidity measured inside the chamber.

When humidity increases above a specified limit:

$$RH > RH_{max}$$

the ventilation fan can be activated.

The fan removes humid air and promotes the movement of moisture away from the crop.

The basic logic is:

IF RH > RHmax

Fan = ON

ELSE

Fan = OFF

In an actual dryer, the fan may also operate continuously or according to a combined temperature-humidity strategy depending on the chamber design.

8.8 Combined Temperature and Humidity Control

A more effective control strategy considers both temperature and humidity.

For example:

Table 8.1.Temperature and humidity based control logic

Temperature Condition	Humidity Condition	Heater	Fan
Low	Low	ON	Normal
Normal	Normal	Maintain	Normal
High	Normal	OFF	ON
Normal	High	Maintain	ON
High	High	OFF	ON
Critical	Any	OFF	ON

This strategy allows the dryer to respond to multiple environmental conditions.

The general control logic is:

$$T < T_{low} \Rightarrow Heater = ON$$

$$T > T_{high} \Rightarrow Heater = OFF$$

$$RH > RH_{max} \Rightarrow Fan = ON$$

$$T > T_{safe} \Rightarrow Heater = OFF, Fan = ON$$

where T_{safe} represents the maximum permitted safety temperature.

8.9 Automatic Heater Control

The heater is controlled through a relay, solid-state relay, or another suitable switching circuit.

The sequence is:

1. Temperature sensor measures chamber temperature.
2. Microcontroller receives the measurement.
3. Temperature is compared with the lower threshold.
4. If temperature is too low, heater is activated.
5. Temperature continues to be monitored.
6. When the upper threshold is reached, heater is deactivated.
7. The process repeats.

The objective is not simply to maximize temperature but to maintain an appropriate drying environment.

8.10 Automatic Fan Control

The fan has two primary functions:

- Air circulation
- Removal of humid air

The fan can be controlled based on humidity and temperature.

For example:

$$RH > RH_{max} \Rightarrow Fan = ON$$

and:

$$T > T_{high} \Rightarrow Fan = ON$$

Thus, the fan may operate when either excessive humidity or excessive temperature is detected. In some implementations, the circulation fan may be kept continuously ON while a separate exhaust fan is controlled according to humidity.

8.11 Safety Control Algorithm

Safety control is an essential part of an automated dryer.

The system should have a maximum safety temperature:

$$T_{safe}$$

If:

$$T \geq T_{safe}$$

the heater must be switched OFF immediately.

The fan may remain ON to assist with cooling, depending on the design.

The safety algorithm can be represented as:

IF T >= Tsafe

Heater = OFF

Fan = ON

Safety Alarm = ON

ELSE

Normal Control

This safety mechanism reduces the possibility of overheating caused by control errors or unexpected conditions. For a practical system, hardware-level thermal protection should also be considered because software alone should not be the only protection against overheating.

8.12 Sensor Failure Detection

Sensor failure can result in incorrect control decisions. Therefore, sensor readings should be checked before they are used.

For example, if a temperature sensor produces an impossible value, the controller can identify the condition as a fault.

A generalized check is:

$$T_{min,valid} \leq T \leq T_{max,valid}$$

If the measured value falls outside the valid range, the system can enter a safe state.

For example:

IF Sensor Error

Heater = OFF

Fan = Safe State

Fault = TRUE

Send Alert

The exact fault-handling strategy depends on the sensor and hardware architecture.

CHAPTER – IX

RESULTS AND PERFORMANCE ANALYSIS

CHAPTER - IX

RESULTS AND PERFORMANCE ANALYSIS

9.1 Introduction

The performance evaluation of the proposed IoT-Based Smart Crop Dryer was carried out to determine the effectiveness of the developed system in monitoring and controlling the crop-drying process. The main purpose of the experimental analysis was to evaluate the ability of the system to maintain suitable drying conditions, reduce moisture content, automate heater and fan operation, and provide real-time monitoring through the IoT platform.

The performance of a crop dryer depends on several parameters, including drying temperature, relative humidity, airflow, initial crop moisture content, final moisture content, drying duration, energy consumption, and environmental conditions.

In the proposed system, temperature and relative humidity were continuously measured using sensors installed in the drying chamber. The microcontroller processed the measurements and operated the heater and fan according to the control algorithm described in Chapter 6. The sensor readings were also transmitted to the IoT monitoring platform.

The experimental results can be used to evaluate the following performance parameters:

- Temperature regulation
- Relative humidity variation
- Crop moisture reduction
- Drying time
- Heater operation
- Fan operation
- Energy consumption
- Sensor accuracy
- IoT communication performance
- Automation performance

- Comparison with conventional drying

9.2 Experimental Objectives

The main objectives of the experimental investigation were:

1. To determine whether the dryer can maintain the required temperature range.
2. To monitor humidity variation during drying.
3. To determine the reduction in crop moisture content.
4. To determine the total drying time.
5. To evaluate automatic heater control.
6. To evaluate automatic fan control.
7. To determine the energy consumed during drying.
8. To evaluate sensor performance.
9. To verify IoT-based remote monitoring.
10. To compare the smart dryer with conventional drying methods.

9.3 Experimental Setup

The experimental setup consists of the drying chamber, heating element, ventilation fan, temperature and humidity sensors, microcontroller, relay/driver circuit, power supply, and IoT monitoring platform. The crop sample is placed inside the drying chamber on the drying trays. The sensors are positioned at suitable locations to measure the environmental conditions surrounding the crop. The microcontroller continuously receives sensor measurements and executes the control algorithm.

9.4 Experimental Procedure

The experimental procedure should be performed under controlled and repeatable conditions.

Step 1: Preparation of Crop

A known quantity of the selected crop is prepared.

The initial mass is recorded as:

$$M_i = \text{Initial crop mass}$$

The initial moisture content is also determined using the selected measurement method.

Step 2: Loading

The crop is evenly distributed on the drying trays.

Uniform distribution is important because it allows air to circulate around the crop.

Step 3: System Initialization

The microcontroller, sensors, IoT communication system, heater, and fan are switched ON.

Step 4: Sensor Verification

Temperature and humidity measurements are checked before beginning the drying process.

Step 5: Drying

The automatic control system is activated.

The controller continuously measures:

$$T(t)$$

and

$$RH(t)$$

where t represents drying time.

Step 6: Data Recording

Temperature, humidity, actuator status, and other relevant parameters are recorded at predefined intervals.

Step 7: Moisture Measurement

Crop moisture content or crop mass is measured at regular intervals.

Step 8: Drying Completion

The drying process is stopped when the selected final moisture condition is reached.

Step 9: Final Measurement

The final crop mass and moisture content are recorded.

9.5 Experimental Parameters

The following parameters should be included in the experimental report.

Table 9.1.Parameters and symbols used in dryer performance analysis

Parameter	Symbol	Unit
Initial crop mass	M_i	kg
Final crop mass	M_f	kg
Initial moisture content	MC_i	%
Final moisture content	MC_f	%
Drying temperature	T	°C
Relative humidity	RH	%
Drying time	t_d	h
Heater power	P_H	W
Fan power	P_F	W
Energy consumption	E	kWh

The actual values should be obtained from the experimental setup.

9.6 Temperature Analysis

Temperature is one of the most important parameters affecting the drying rate.

During the initial stage of operation, the chamber temperature increases because thermal energy is supplied by the heating element.

As the temperature approaches the set point, the control algorithm switches the heater OFF or maintains its state according to the hysteresis control strategy.

A typical temperature profile may have three stages:

Stage 1 – Heating Stage

The chamber temperature increases rapidly.

$$T(t) \uparrow$$

Stage 2 – Controlled Drying Stage

The temperature fluctuates around the desired operating range.

$$T_{low} \leq T(t) \leq T_{high}$$

Stage 3 – Final/Cooling Stage

After drying is completed, the heater is switched OFF and the chamber is allowed to cool under controlled conditions.

9.7 Temperature Analysis

Temperature is one of the most important parameters affecting the drying rate.

During the initial stage of operation, the chamber temperature increases because thermal energy is supplied by the heating element.

As the temperature approaches the set point, the control algorithm switches the heater OFF or maintains its state according to the hysteresis control strategy.

A typical temperature profile may have three stages:

Stage 1 – Heating Stage

The chamber temperature increases rapidly.

$$T(t) \uparrow$$

Stage 2 – Controlled Drying Stage

The temperature fluctuates around the desired operating range.

$$T_{low} \leq T(t) \leq T_{high}$$

Stage 3 – Final/Cooling Stage

After drying is completed, the heater is switched OFF and the chamber is allowed to cool under controlled conditions.

between successive measurements:

$$DR_i = \frac{MC_i - MC_{i+1}}{t_{i+1} - t_i}$$

This allows the drying rate to be plotted against time.

9.8 Energy Consumption Analysis

Energy consumption is an important parameter because heating is usually one of the major energy-consuming operations in a dryer.

For an electrical load:

$$E = P \times t$$

where:

- E = energy consumed

- P = power
- t = operating time

For energy in kWh:

$$E_{kWh} = \frac{P_W t_h}{1000}$$

If both heater and fan energy are considered:

$$E_{total} = E_H + E_F + E_C$$

where:

- E_H = heater energy
- E_F = fan energy
- E_C = controller and electronics energy

If the actual power consumption is measured using a power meter, the measured value should be preferred over a theoretical calculation.

9.9 Specific Energy Consumption

Specific energy consumption can be calculated to compare the efficiency of different drying trials.

One possible definition is:

$$SEC = \frac{E_{total}}{M_w}$$

where:

- SEC = specific energy consumption
- E_{total} = total energy consumed
- M_w = mass of water removed

The unit can be expressed as:

$$kWh/kg\ water$$

This provides a more meaningful comparison between trials with different crop quantities.

9.10 Temperature Control Performance

The effectiveness of the automatic controller can be evaluated by examining how closely the actual temperature follows the desired set point.

The following parameters can be calculated:

Average Temperature

$$T_{avg} = \frac{\sum T_i}{n}$$

Maximum Temperature

$$T_{max} = \max(T_i)$$

Minimum Temperature

$$T_{min} = \min(T_i)$$

Temperature Variation

$$\Delta T = T_{max} - T_{min}$$

Average Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n} | T_i - T_{set} |$$

A lower MAE indicates closer tracking of the desired temperature.

9.11 Temperature Overshoot

The temperature overshoot can be calculated using:

$$\%OS = \frac{T_{max} - T_{set}}{T_{set}} \times 100$$

If the maximum temperature is close to the set point, the system demonstrates better temperature regulation.

The experimental temperature graph should be used to identify the maximum temperature.

9.12 Comparison With Conventional Drying

The proposed smart dryer can be compared with conventional open-air or manually controlled drying.

Possible comparison parameters include:

Table 9.2.Performance and control features of conventional and IOT smart drying systems

Parameter	Conventional Drying	IoT Smart Dryer
Temperature monitoring	Manual/limited	Automatic
Humidity monitoring	Limited	Continuous
Heater control	Manual/Not applicable	Automatic
Fan control	Manual/Not applicable	Automatic
Remote monitoring	No	Yes
Data logging	Limited	Possible
Manual intervention	Higher	Lower
Environmental dependency	High	Lower
Drying consistency	Variable	More controlled

The exact comparison should be based on the method actually used in your project.

The chamber temperature increases rapidly.

$$T(t) \uparrow$$

Stage 2 – Controlled Drying Stage

The temperature fluctuates around the desired operating range.

$$T_{low} \leq T(t) \leq T_{high}$$

Stage 3 – Final/Cooling Stage

After drying is completed, the heater is switched OFF and the chamber is allowed to cool under controlled conditions.

REFERENCES

REFERENCES

[1] M. M. Mkhize, "Review of solar crop drying efficiency and its interconnection with meteorological factors," *Discover Energy*, vol. 5, Art. no. 22, 2025, doi: 10.1007/s43937-025-000882.

[2] K. Venkateswarlu and S. V. Kota Reddy, "Recent trends on energy-efficient solar dryers for food and agricultural products drying: A review," *Waste Disposal & Sustainable Energy*, vol. 6, pp. 335–353, 2024, doi: 10.1007/s42768-024-00193-3.

[3] J. I. Miano *et al.*, "Optimizing drying efficiency through an IoT-based direct solar dryer system: Integration of web data logger and SMS notification," *International Journal of Advanced Computer Science and Applications (IJACSA)*, vol. 14, no. 7, pp. 233–240, 2023.

[4] P. R. Patil, Rakshith, T. J. Kotian, and Y. G. T., "IoT-based areca nut dryer," *International Journal of Engineering Research & Technology (IJERT)*, vol. 13, no. 11, Nov. 2024, Paper ID: IJERTV13IS110090.

[5] A. Welankiwar *et al.*, "IoT based solar dryer for agriculture products," *International Journal of Innovative Science and Research Technology (IJISRT)*, vol. 10, no. 11, pp. 2475–2481, Nov. 2025, doi: 10.38124/ijisrt/25nov1349.

[6] A. A. Michael, U.-O. D. Chukunebikpe, A. A. Omowale, and F. A. Olutosin, "Development of an Internet of Things (IoT)-based solar chilli pepper dryer," *Dutse Journal of Pure and Applied Sciences (DUJOPAS)*, vol. 10, no. 4a, pp. 129–138, Dec. 2024, doi: 10.4314/dujopas.v10i4a.13.